



ANSWERS TO THE PROBLEMS

“The researcher suffers disappointments, long months spent in the wrong direction, failures. But failures are also useful, because, properly analyzed, they can lead to success. And for the researcher, there is no joy comparable to that of a discovery, no matter how small....”

Alexander Fleming



Ministry of Education
Republic of Cuba

National Chemistry Contest - Cuba 2025**Answers to the problems****Authors**

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“Dear contestants, we work for you because you are a source of pride for family, friends, teachers, administrators, and for Cuban education. On this occasion, we decided to dedicate each of the questions to some of the students and mentors from last year's Pre-Selection: Héctor Manuel, Mario Luis, Juan Carlos, Roy Roger, Marco Antonio, Ernesto Alejandro, Osmani Lázaro, Melissa María, Bárbaro Eriel, Pedro Alexis, Renny Alejandro, Cynthia María, Leticia María, Carlos Daniel, Lauren, Carlos Manuel, Ian David, and Sergio Daneel; and to some provincial mentors: Rolando, Orestes, Julio, and Agustín.”

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Free distribution

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Problem distribution

	First Day of Exams						Second Day of Exams					
	P1 (10%)	P2 (15%)	P3 (10%)	P4 (20%)	P5 (15%)	P6 (15%)	P7 (15%)	P8 (10%)	P9 (15%)	P10 (15%)	P11 (20%)	P12 (20%)
10th	X	X	X	X	---	---	X	X	X	X	---	---
11th	X	X	X	---	X	---	X	X	X	---	X	---
12th	X	X	---	---	X	X	X	---	---	---	X	X

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Sphere volume	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Sphere area	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

Specific instructions for the Chemistry Contest

Specific instructions for the Chemistry Contest:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information.** Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables.
No questions will be canceled due to poor printing.
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field;** it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the first day,
 - 10th grade should answer problems 1, 2, 3 and 4.
 - 11th grade should answer problems 1, 2, 3 and 5.
 - 12th grade should answer problems 1, 2, 5 and 6.
- **Check that your exam is complete.** In the case of the exam on the second day,
 - 10th grade should answer problems 7, 8, 9 and 10.
 - 11th grade should answer problems 7, 8, 9 and 11.
 - 12th grade should answer problems 7, 11 and 12.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation.
- Attention! You have limited time and the exam has a large number of items.
- You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.
- The final grade will be rounded to the nearest unit for the student’s advantage in all cases.

You are a source of pride for Cuban education: Best of luck!

Problem 1. Hydrides (10% of total; 10th, 11th and 12th grades)

1.1	1.2	1.3	1.4	1.5	1.6	Total	Total
11	7	10	2	5	5	40	10%
11	7	10	2	5	5	40	10%

One day, Lauren and Leticia were preparing a lesson about hydrides for tenth-grade students. Lauren wrote the following statement. Can you fill in the blanks to make it understandable?

1.1. Hydrogenated binary compounds can be classified into four main types. i) Ionic/salty hydrides are formed with the least electronegative metals. They have an ionic chemical bond and a large negative partial charge on the hydrogen atom. They react vigorously with water, releasing hydrogen, which reflects their potent basic and reducing characteristics. ii) Interstitial/metallic hydrides are formed when hydrogen atoms occupy the empty spaces in the crystal lattices of the transition elements. They have chemical properties like the previous ones, although more attenuated, because in them the partial charge on the hydrogen atom is less negative. They are electrical conductors in the solid state because the chemical bond they possess is metallic. iii) Polymeric/intermediate hydrides are formed with some chemical elements from groups 2, 11, 12, and 13, which have lower electronegativities and empty orbitals. The chemical bond in this case has intermediate characteristics. These hydrides react readily with the non-bonding electron pairs of Lewis bases. iv) Volatile hydrides are formed with non-metals and their acid-base behavior varies. Group 15 hydrides behave as bases because they have a non-bonding electron pair in their molecular structure. For example, ammonia in aqueous solution increases the concentration of hydroxide ions in water. In contrast, group 16 and 17 hydrides in aqueous solution behave as acids because they have a large partial positive charge on the hydrogen atom, thereby increasing the concentration of hydronium ions in water. They react vigorously with alkali metals, releasing hydrogen, which reflects their acidic and oxidizing characteristics. Hydrides of the first three types are solids at room temperature, while most type iv hydrides are gases, with water being a notable exception. Hydrogenated binary compounds of carbon, known as hydrocarbons, can exist in all three aggregation states: as the chain length increases, the melting and boiling points increases due to an increase in dispersion/London/Van der Waals interactions.

11 pt (0,5 pt each)

For her part, Leticia found that ammonia has anomalous properties compared to the other hydrides in group 15, as shown in the following table.

Hydride	NH ₃	PH ₃	AsH ₃	SbH ₃	BiH ₃
T _{eb} (°C)	-33	-88	-63	-17	17
α H-X-H (°)	107	93	92	92	91

1.2. Explain in detail why

i) The boiling point of ammonia is unusually high.

Ammonia has a free electron pair and high bond polarity, allowing it to form hydrogen bonds between its molecules, which increases its boiling point. **3 pt**

ii) The bond angle is close to 90° in the hydrides mentioned above, except for ammonia.

In the heavier hydrides, the bond is formed between the 1s orbitals of hydrogen and the very large p orbitals of the nonmetal, which are perpendicular to each other. **2 pt**

On the contrary, in ammonia, the bond forms between the 1s orbitals of hydrogen and the 2sp³ orbitals of nitrogen. The similar size and energy of the 2s and 2p orbitals of nitrogen allow them to mix effectively to form hybrid orbitals. **2 pt**

Leticia and Lauren decided to demonstrate a practical application of a volatile hydride. Under a fume hood, they set up the equipment shown in the figure to generate a certain toxic and odorous gas *in situ*, which they then bubbled into 1,0 mol/L aqueous nitrate solutions. The results obtained were as follows:

$\text{Al}(\text{NO}_3)_3$: No change was observed.

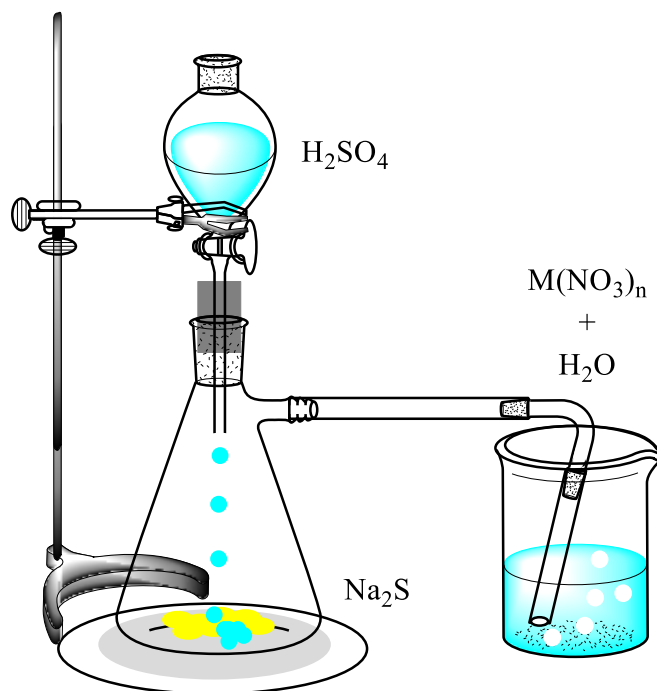
$\text{Fe}(\text{NO}_3)_3$: A black precipitate and a yellowish-white turbidity formed in the solution.

$\text{Co}(\text{NO}_3)_2$: A black precipitate formed and the pink solution became discolored.

$\text{Cu}(\text{NO}_3)_2$: A black precipitate formed and the blue solution became discolored.

$\text{Pb}(\text{NO}_3)_2$: A black precipitate formed.

All the final solutions became acidic, although the first was the least acidic of all.



1.3. Formulate balanced chemical equations (ionic or molecular) corresponding to all reactions that occurred in the procedures described. Indicate the states of aggregation.



1.4. Why are Fe^{3+} , Co^{2+} and Cu^{2+} ions colored in aqueous solution?

Fe^{3+} , Co^{2+} and Cu^{2+} ions have the 3d sublevel partially occupied, so electronic transitions to higher levels can occur when absorbing visible light. **2 pt**

1.5. Both contestants were struck by the fact that sulfide anions have a greater affinity for the cations Fe^{2+} , Co^{2+} , Cu^{2+} and Pb^{2+} than for the cations Al^{3+} and Fe^{3+} . Why?

Sulfide anions are soft bases and have a greater affinity for the intermediate acids Fe^{2+} , Co^{2+} and soft acid Pb^{2+} , than for the hard acids Al^{3+} and Fe^{3+} . **5 pt**

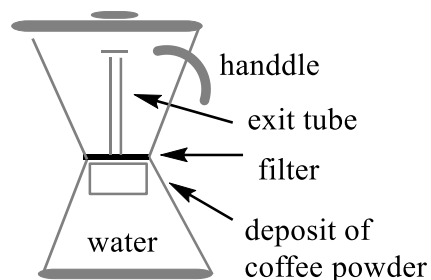
1.6. Identify the laboratory tools shown in the figure.

Universal stand, extension clamp with jaw, separating or dropping funnel, rubber stopper, adapter, rubber/glass tube section, Erlenmeyer flask with side fitting, beaker, asbestos-coated metallic mesh or metallic mesh. **5 pt** (0,5 pt each)

Problem 2. A can of coffee (15% of total; 10th, 11th and 12th grades)

2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	Total	Total
2	2	5	3	18	6	2	10	12	60	15%
2	2	5	3	18	6	2	10	12	60	15%

Most Cubans enjoy drinking coffee. If you visit a home or institution, you'll likely be offered a white porcelain cup containing about 50 mL of the beverage. Depending on the location, coffee can be heated with electricity, natural gas, coal, or firewood. Typically, coffee is prepared in a coffee pot, in an energy-intensive process. First, the water must be brought from room temperature, 25°C, to the boiling point, 100°C, consuming energy amount Q_1 . Then, a small amount of water turns into a gaseous state, causing an increase in pressure inside the lower reservoir and consuming energy amount Q_2 . Liquid water then mixes with coffee powder and extracts the main water-soluble component of coffee, caffeine; the formation of bubbles aids the extraction process. The excess pressure then causes the resulting solution to escape through the outlet tube. Finally, all that remains is to wait for the coffee to cool slightly and enjoy the beverage. A four-cup Cuban coffee pot can contain up to 200 mL of water; however, at the end of a "colada", only about 190 mL of coffee is obtained, which can be attributed primarily to evaporation losses.



To answer all the following questions, consider coffee to be an ideal dilute solution of the solute caffeine in the solvent water. Therefore, the heat capacity and density of coffee are equal to those of pure water, respectively, $c = 4,18 \text{ J}\cdot\text{K}^{-1}\cdot\text{g}^{-1}$ and $\rho = 1,00 \text{ g/mL}$. Also, use the thermodynamic data provided at the end of the question.

2.1. Coffee pots are made of metal, which facilitates both rapid heating and cooling. Why are metals good conductors of heat?

In metals, there are freely moving electrons that effectively transfer thermal energy throughout the material. **2 pt**

2.2. How much energy Q_1 (kJ) is required to heat all the water inside a full coffee pot?

$$Q_1 = m(\text{H}_2\text{O}) \cdot c(\text{H}_2\text{O}) \cdot \Delta T$$

$$Q_1 = 200 \cdot 4,18 \cdot (100 - 25) = 62700 \text{ J} = 62,7 \text{ kJ} \quad \textbf{2 pt}$$

2.3. How much energy Q_2 (kJ) is required to evaporate 10 mL of water?

$$\Delta H_{\text{vap}}^\circ(\text{H}_2\text{O}) = \Delta H_f^\circ(\text{H}_2\text{O}, \text{g}) - \Delta H_f^\circ(\text{H}_2\text{O}, \text{l})$$

$$\Delta H_{\text{vap}}^\circ(\text{H}_2\text{O}) = -242 - (-286) = 44 \text{ kJ/mol} \quad \textbf{2 pt}$$

$$M(\text{H}_2\text{O}) = 18 \text{ g/mol} \quad \textbf{1 pt}$$

$$Q_2 = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} \cdot \Delta H_{\text{vap}}^\circ(\text{H}_2\text{O}) = \frac{10}{18} \cdot 44 = 24 \text{ kJ} \quad \textbf{2 pt}$$

2.4. What amount (mole) of water vapor exists inside the lower container when the last portion of the liquid is coming out? Assume that water vapor behaves in an ideal manner.

The gas pressure is approximately equal to the atmospheric pressure. **1 pt**

$$n(\text{H}_2\text{O}, \text{vap}) = \frac{P \cdot V}{R \cdot T} = \frac{101,3 \cdot 0,200}{8,314 \cdot 373,15} = 0,00653 \text{ mol} \quad \textbf{2 pt}$$

2.5. is the minimum amount of electricity (moles of electrons), natural gas (L at standard ambient conditions), coal (kg), and wood (kg) required to brew coffee in a coffee pot 1000 times?

$$Q_1 + Q_2 = 62,7 + 24 \approx 86,7 \text{ kJ} \quad \underline{2 \text{ pt}}$$

$$Q_{\text{total}} = 1000 \cdot 86,7 = 86700 \text{ kJ} \quad \underline{2 \text{ pt}}$$

$$n(\text{elect}) = \frac{Q_{\text{total}}}{\Delta V \cdot e \cdot N_A} = \frac{86700 \cdot 10^3}{110 \cdot 1,609 \cdot 10^{-19} \cdot 6,022 \cdot 10^{23}} = 8,27 \text{ mol} \quad \underline{2 \text{ pt}}$$



$$\Delta H_{\text{comb}}^\circ(\text{CH}_4) = -\Delta H_f^\circ(\text{CH}_4) + \Delta H_f^\circ(\text{CO}_2) + 2\Delta H_f^\circ(\text{H}_2\text{O}, \text{l})$$

$$\Delta H_{\text{comb}}^\circ(\text{CH}_4) = -(-79) - 394 - 2 \cdot 286 = -887 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

$$V(\text{CH}_4) = \frac{Q_{\text{total}}}{\Delta H_{\text{comb}}^\circ(\text{CH}_4)} \cdot V_m = \frac{86700}{887} \cdot 24,8 = 2424 \text{ L} \quad \underline{2 \text{ pt}}$$

$$\Delta H_{\text{comb}}^\circ(\text{C}) = \Delta H_f^\circ(\text{CO}_2) = -394 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

$$m(\text{C}) = \frac{Q_{\text{total}} \cdot A_r(\text{C})}{\Delta H_{\text{comb}}^\circ(\text{C}) \cdot 1000} = \frac{86700 \cdot 12,0}{394 \cdot 1000} = 2,64 \text{ kg} \quad \underline{2 \text{ pt}}$$

$$m(\text{wood}) = \frac{Q_{\text{total}}}{\Delta H_{\text{comb}}^\circ(\text{wood}) \cdot 1000} = \frac{86700}{18 \cdot 1000} = 4,8 \text{ kg} \quad \underline{2 \text{ pt}}$$

In the village of Jatibonico, mornings are cold, typically around 15 °C. Mario enjoys drinking coffee in the morning, but in his village, the beverage must be insulated in a thermos to keep it hot for a long time. The organizers gave all participants of the 56th ICHO a 1,0 L thermos, which Mario uses to store coffee. The heat flux Q/t can be calculated using the following equation: where k is the thermal conductivity coefficient of the material, A is the cross-sectional area through which heat energy transfer occurs, l is the thickness of the material, and ΔT is the temperature difference.

$$\frac{Q}{t} = \frac{k \cdot A \cdot \Delta T}{l}$$

2.6. Estimate the minimum time t (in min) required for all the freshly brewed coffee inside Mario's thermos to cool down to 15 °C. Consider the cylindrical thermos (diameter 10,0 cm; height 14,0 cm) made of a material with $k = 0,15 \text{ W} \cdot \text{m}^{-1} \cdot ^\circ\text{C}^{-1}$ and a thickness of 0,50 cm.

$$A_{\text{termo}} = 2\pi rh + 2\pi r^2 = 2 \cdot \pi \cdot 5 \cdot 14 + 2 \cdot \pi \cdot 5^2 = 597 \text{ cm}^2 = 5,97 \cdot 10^{-2} \text{ m}^2 \quad \underline{2 \text{ pt}}$$

The initial cooling rate will be

$$\frac{Q}{t} = \frac{k \cdot A \cdot \Delta T}{l} = \frac{0,15 \cdot 5,97 \cdot 10^{-2} \cdot (100 - 15)}{0,005} = 152,24 \frac{\text{J}}{\text{s}} = 0,152 \frac{\text{kJ}}{\text{s}} \quad \underline{1 \text{ pt}}$$

The heat needed to cool all the coffee inside the 1 L thermos will be

$$Q_{\text{cooling}} = m(\text{coffee}) \cdot c(\text{coffee}) \cdot \Delta T = 1000 \cdot 4,18 \cdot (100 - 15) = 355300 \text{ J} = 355 \text{ kJ} \quad \underline{2 \text{ pt}}$$

$$t = \frac{Q_{\text{cooling}}}{Q/t} = \frac{355}{0,152} = 2338 \text{ s} \approx 39 \text{ min} \quad \underline{1 \text{ pt}}$$

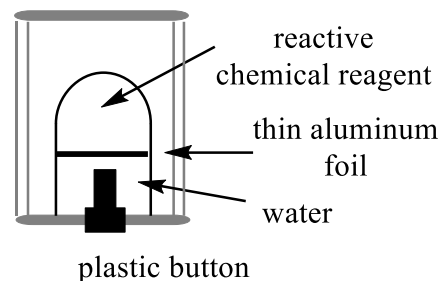
It is acceptable to calculate using 190 mL of coffee, in which case $t = 7,4 \text{ min}$.

2.7. In practice, the cooling time is longer than you calculated. Why?

As heat is transferred to the medium, the temperature difference decreases and the cooling rate decreases.
 2 pt

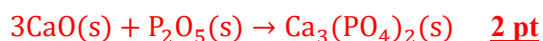
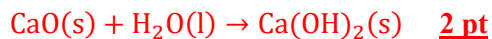
Emmanuel and *Riky* live in the city, but they enjoy hiking, biking, and overnight camping. Self-heating containers are more useful than thermoses. By pressing a button, they can break a film, separating a suitable solid chemical from the water in a small reservoir inside the container. An exothermic process then occurs, heating the entire container. The solid can be composed of a pure substance or a mixture, for example,

CaO, CaCl₂, Mg + Fe, AlCl₃ + CaO, P₂O₅ + CaO, CuSO₄ + Zn.



2.8. Formulate the equations corresponding to the reactions that occur (if any)

in each case. In the third formulation, the iron acts as a redox catalyst; it is not consumed in the process.



No reaction occurs with CaCl₂, but the heating is due to the dissolution of the salt.

2.9. One of the products formed, calcium hydroxide, has a high enthalpy of formation, which makes the process exothermic. Estimate the value of this magnitude from the corresponding thermodynamic data. Use the Born-Haber cycle and the Kapustinskii equation to calculate the lattice energy.

$$U(\text{Ca(OH)}_2) \approx -1,20 \cdot 10^5 \cdot \frac{3 \cdot 2 \cdot 1}{114 + 133} \cdot \left(1 - \frac{34,5}{114 + 133}\right) = -2508 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

$$\Delta H_f^\circ(\text{Ca(OH)}_2) = \Delta H_{\text{sub}}(\text{Ca}) + I_1(\text{Ca}) + I_2(\text{Ca}) + E(\text{O}=\text{O}) + E(\text{H}-\text{H}) - 2 \cdot E(\text{O}-\text{H}) + 2 \cdot \Delta H_{\text{EA}}(\text{O}-\text{H}) + U(\text{Ca(OH)}_2) \quad \underline{8 \text{ pt}}$$

$$\Delta H_f^\circ(\text{Ca(OH)}_2) = 178 + 590 + 1145 + 495 + 434 - 2 \cdot 465 + 2 \cdot (-177) - 2508$$

$$\Delta H_f^\circ(\text{Ca(OH)}_2) = -950 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

The experimental value reported in the literature is -985 kJ/mol.

Useful data:

In Cuba, the electricity grid supplies direct current with a voltage of 110 V. Natural gas is composed mostly of methane. The average heat of combustion of wood is 18 kJ/g.

$$\Delta H_f^\circ(\text{X})/\text{kJmol}^{-1}: \text{H}_2\text{O,liquid} = -286; \text{H}_2\text{O,vapor} = -242; \text{CH}_4 = -79; \text{CO}_2 = -394;$$

The charge of an electron is $e = 1,602 \cdot 10^{-19} \text{ C}$; one Faradio is 1 mol of electrons.

The fundamental equation of calorimetry is $Q = m \cdot c \cdot \Delta T$, where c is the heat capacity of the material.

The following values are expressed in [kJ/mol]: $I_1(\text{Ca}) = 590$, $I_2(\text{Ca}) = 1145$, $\Delta H_{\text{EA}}(\text{OH}) = -177$ (electron affinity of the hydroxyl radical), $E(\text{O}=\text{O}) = 495$, $E(\text{H}-\text{H}) = 434$, $E(\text{O}-\text{H}) = 465$, $\Delta H_{\text{sub}}(\text{Ca}) = 178$.

Kapustinskii's equation, U is the lattice enthalpy in kJ/mol, r and z are the radii in pm and the charges of the ions, respectively, $\sum v$ is the total number of ions in the empirical formula, $\sum v = 3$ for Ca(OH)_2 .

$$U \approx -1,20 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+||z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

The thermochemical radii of calcium and hydroxide ions are 114 and 133 pm, respectively.

Problem 3. Electronegativity (10% of total; only 10th and 11th grades)

3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	3.10	3.11	3.12	Total	Total
2	4	2	4	2	2	3	6	2	6	3	4	40	10%
2	4	2	4	2	2	3	6	2	6	3	4	40	10%

In the Pre-Selection, there are two students named Carlos, who come from different provinces: one from Sancti Spiritus and the other from Camagüey. They are friends and don't mind the historical rivalry between their teams. One day, they decided to study the theoretical foundations of chemical bonding together, but they stumbled upon the obscure concept of "electronegativity", and a heated debate ensued. Electronegativity occupies a central position in describing chemical bonding, and since it was postulated by Linus Pauling in 1932, it has been a fundamental tool for understanding charge transfer, bond polarity and strength, reactivity, and various chemical properties.

3.1. the concept of electronegativity.

It is the average force of attraction exerted by the nucleus on the valence electrons.

Alternative definitions are accepted. **2 pt**

3.2. Explain how this property varies across the periodic table and the causes of the behavior described.

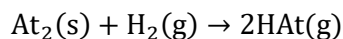
In a period, the increase in effective nuclear charge and the contraction of atoms cause electronegativity to increase as the atomic number increases. In a group, the increase in atomic radius causes electronegativity to decrease as the atomic number increases, because the attraction of the nucleus to the valence electrons decreases. **4 pt**

Although electronegativity χ cannot be measured, as it is not a physical quantity, it can be estimated indirectly. Carlos Daniel found the original definition used by Pauling, based on thermodynamic data of bond enthalpies (in kJ/mol) and taking hydrogen as an arbitrary reference, $\chi_H = 2,20$.

$$X_A - X_B = 0,102 \cdot \sqrt{\Delta H(A-B) - \frac{\Delta H(A-A) + \Delta H(B-B)}{2}}$$

According to this definition, the additional stability of a covalent bond between two elements is due to the charge transfer or ionic contribution resulting from the difference in electronegativities.

3.3. Carlos Manuel found a periodic table and noticed that astatine has the same electronegativity as dihydrogen. What is the enthalpy change for the following reaction?



Since there is no ionic contribution to the bond formed, the enthalpy change of the reaction is approximately zero. **2 pt**

3.4. So, Carlos Daniel decided to calculate the electronegativity values of the remaining halogens to verify that they matched those in the table. What values did he obtain? $\Delta H^\circ(\text{A-B})/\text{kJ/mol}$: H-H = 436, F-F = 156, Cl-Cl = 243, Br-Br = 193, I-I = 151, H-F = 569, H-Cl = 431, H-Br = 368, H-I = 297.

$$X_F = 2,20 + 0,102 \cdot \sqrt{569 - \frac{156 + 436}{2}} = 3,89$$

$$X_{Cl} = 2,20 + 0,102 \cdot \sqrt{431 - \frac{243 + 436}{2}} = 3,18$$

$$X_{\text{Br}} = 2,20 + 0,102 \cdot \sqrt{368 - \frac{193 + 436}{2}} = 2,95$$

$$X_{\text{I}} = 2,20 + 0,102 \cdot \sqrt{297 - \frac{151 + 436}{2}} = 2,39$$

4 pt (1 pt each)

3.5. The binding energy of fluorine is unusually low. What is the reason for this anomaly?

In fluorine, there is a strong electrostatic repulsion due to the high electron density that weakens the bond. / The antibonding effect of the σ_u^* orbital is stronger in fluorine due to the greater overlap of the orbitals resulting from the small bond distance. **2 pt**

Carlos Manuel decided to apply the concept of electronegativity to explain the origin of bond polarity. When two atoms bond, an electron transfer occurs from the less electronegative atom to the more electronegative atom. Consequently, a partial charge δ , of equal magnitude but different signs, is generated on each atom, and the bond is said to be polar. The dipole moment μ of a molecule, which in the case of diatomic molecules, is a direct measure of bond polarity, can be experimentally determined using the formula $\mu = \delta \cdot d$, where d is the separation between the charges, in the case analyzed, the bond distance.

Molecule	H-F	H-Cl	H-Br	H-I	H-At
μ (D)	1,86	1,11	0,79	0,38	~ 0

3.6. Explain the variation in the values that appear in the table above to Carlos Manuel.

As the electronegativity of the halogen decreases, the electronegativity difference decreases, and consequently, the partial charge declines as well. This effect predominates over the increase in atomic radius, and the net result is a progressive decrease in the dipole moment. **2 pt**

3.7. Carlos Daniel discovered that during bond formation, the electronegativity of the atoms varies until the electronegativities of all the atoms are equal, according to Sanderson's Principle. Can you explain the origin of this variation and which atom increases/decreases electronegativity?

The more electronegative atom in isolation receives electrons and becomes negatively charged, so it loses some of its ability to gain more electrons (it repels them more), and its electronegativity decreases. The less electronegative atom in isolation donates electrons and becomes positively charged, so its ability to gain more electrons (attracts them) increases slightly. **3 pt**

3.8. A Lewis acid is a substance with an empty orbital that can accept electrons, while a Lewis base is a substance with a filled orbital that can donate electrons. Help Carlos Manuel to identify which of the following pairs of substances are Lewis acids or bases, and specify which substance is the stronger acid or base for each pair; explain your answer.

i) $\text{H}_2\text{O}/\text{H}_3\text{N}$, ii) BCl_3/BI_3 , iii) $\text{HO}^-/\text{H}_2\text{O}$, iv) Cl_2/I_2

NH_3 is a better base, because the nitrogen atom is less electronegative.

BCl_3 is a better acid, because it has a higher δ^+ as the difference in electronegativity is greater.

HO^- is a better base, because it has a net negative charge.

I_2 It is a better acid, because it has the lowest energy σ^* orbital and the bond is weaker.

6 pt (1,5 pt each, 0,5 pt for identifying and 1 pt for explaining, 0 pt for classifying in acid/base)

An oxide or hydroxide is more basic the greater the partial negative charge on the oxygen atom. Oxides or hydroxides are more acidic the greater the partial positive charge on the central and hydrogen atoms. The acid-base character of oxides and hydroxides varies regularly across the periodic table, depending on the oxidation number and the electronegativity of the atoms. Consider the element chromium, which forms a basic oxide, CrO, an amphoteric oxide, Cr₂O₃, and an acidic oxide, CrO₃.

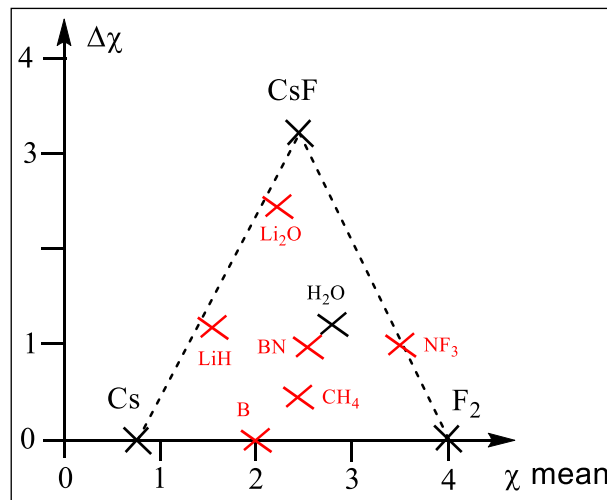
3.9. Why does an increase in the oxidation number cause an increase in the acidity of the oxides?

When the oxidation number is high, there are a greater number of strongly electronegative oxygen atoms surrounding and subtracting electron density from the central atom of the oxide. Consequently, the partial positive charge on the central atom is greater. **2 pt**

With the data for each atom, Carlos Daniel decided to draw a Van Arkel-Keteelar diagram, graphing the difference in electronegativity *versus* the average electronegativity for some substances.

atom	Cs	Li	B	H	C	N	O	F
χ	0,8	1,0	2,0	2,2	2,6	3,0	3,4	4,0

3.10. Find the coordinates of the substances B, BN, NF₃, CH₄, Li₂O and LiH, and place these substances in the appropriate places on the diagram. Indicate for each the predominant type of chemical bond (metallic, ionic, nonpolar covalent, polar covalent). Mark the exact position of each substance with an X and write its formula. Consider the examples provided in the figure (Cs, CsF, F₂ and H₂O).



Ionic: LiH and Li₂O, Polar covalent: NF₃, BN and CH₄, Apolar covalent: B. **3 pt**

3 pt for the position of the substances on the graph (0,5 pt each).

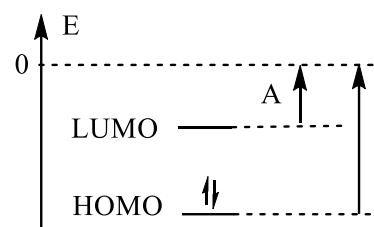
3.11. When Carlos Manuel looked at the graph, he immediately realized the relationship between the three extreme models of chemical bonding (ionic, covalent, and metallic) and electronegativity, and he took note of it. Can you complete the following statement?

When atoms of similar but low electronegativity combine, the resulting bond is metallic, for example, in metals and alloys. When atoms of similar but high electronegativity combine, the bond is predominantly covalent, as is the case in nonmetals and interhalogens/hydrides/etc. On the other hand, atoms of very different electronegativity are held together by ionic bonding, as in salts.

3 pt (0,5 each)

At the end of the study session, students learned that it is also possible to calculate the electronegativity of a substance using the Mulliken method, from the electronic structure, and the electronaffinity A and ionization energy I data.

3.12. How must the values of A and I be like, for electronegativity to be high? Explain in each case.



A must be high, to attract electrons easily. **2 pt**

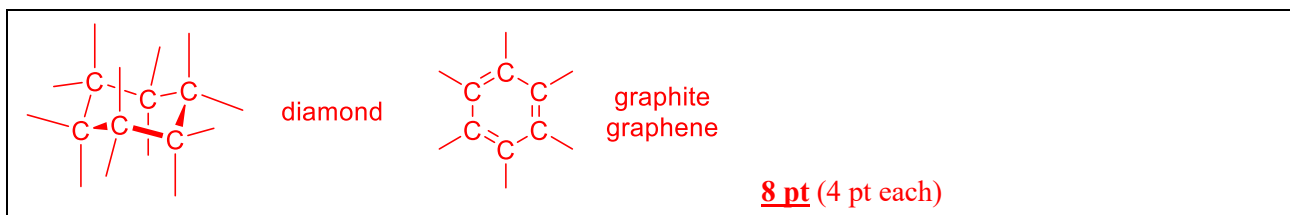
I must be high, to give up electrons with difficulty. **2 pt**

Problem 4. Carbon (20% of total; only 10th grade)

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	Total	Total
8	14	6	3	5	4	10	10	20	80	20%
8	14	6	3	5	4	10	10	20	80	20%

Carbon is a highly versatile and reactive chemical element, a constituent of essential compounds for life and industry. It exists in various allotropic forms, such as diamond, graphite, graphene, nanotubes, and fullerenes, of which graphite is the most stable. These allotropes can be considered to be formed by the repetition of two types of cyclic structural units of six atoms. In diamond, the structural unit is *chair*-shaped, containing only sp^3 -hybridized atoms, spaced 1,54 Å apart. The extension of the structural unit in space generates a three-dimensional network of tetrahedral atoms. In contrast, in graphite and graphene, the structural unit consists of a flat hexagon formed only by sp^2 -hybridized atoms, spaced 1,42 Å apart. The extension of the structural unit in space generates a sheet of only two dimensions (graphene), which can be stacked together to form a three-dimensional network (graphite), with a distance of 3,35 Å between the atoms of different sheets.

4.1. Draw the two structural units described above. Include only the six atoms that comprise them and all the bonds they form between them and other neighboring atoms.



4.2. Explain in detail why

i) Diamond is an electrical insulator, while graphite is a conductor.

In diamond, there are strong σ_{sp^3} covalent bonds between carbon atoms, and the valence electrons are confined to the internuclear region. / As a result of the strong σ_{sp^3} bonds between carbon atoms, the valence (bonding) and conduction (antibonding) bands are widely separated.

In graphite, there are strong σ_{sp^2} covalent bonds between carbon atoms, and $3N$ valence electrons are confined to the internuclear region. However, $1N$ electrons form weak π bonds that are delocalized throughout the atoms in a single layer and can move freely.

4 pt (2 pt each)

ii) Diamond is very hard and is used in metal-cutting tools, while graphite is very soft and is used in powder form as a lubricant for heavy machinery.

The deformation of the diamond structure requires a lot of energy to break the strong σ_{sp^3} covalent bonds between the atoms.

Atoms belonging to different layers in graphite are not bonded to each other but rather interact through weak van der Waals forces. Breaking the crystal structure requires little energy.

4 pt (2 pt each)

iii) The six bond distances in the hexagonal structural unit of graphite are equal.

In graphite all atoms have sp^2 hybridization and contribute equally to the π bonds, thus existing resonance/delocalization. **2 pt**

iv) Properties of graphite vary depending on the direction in which they are measured, a phenomenon known as anisotropy, while the properties of diamond are equal in any direction.

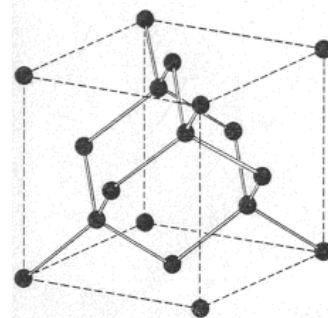
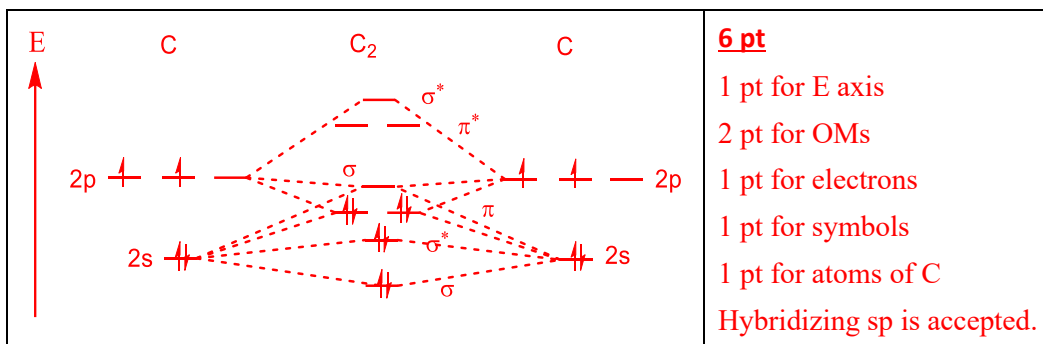
The σ_{sp^3} bonds of diamond are formed between atoms located at the vertices of a tetrahedron, facing equally in all directions of the crystal.

The σ_{sp^2} and π bonds of graphite are formed between the atoms located at the vertices of a hexagon, in a (single) direction parallel to the sheet that forms the solid.

4 pt (2 pt each)

Osmani and Renny are very curious students, and during a bibliographic search, they discovered that carbon can also be present as diamagnetic diatomic molecules in the gaseous state.

4.3. Draw the molecular orbital energy diagram for the C_2 molecule. Note that the 2s and 2p orbital energies are similar, so s-p mixing exists.



Renny wants to study Designing, and because of this, he is deeply drawn to the study of the structure of substances. He knows that the regular structure of diamond can also be divided into small imaginary cubes with an edge length of 3,57 Å. The figure shows one of these cubes, also called a unit cell: there is a carbon atom at each vertex and in the center of each face, as well as an atom at four interior positions of the cube known as tetrahedral sites.

4.4. Explain why each unit cell can be considered to contain a total of eight carbon atoms.

The 8 atoms occupying the vertices are shared between 8 cells, so each cell contains 1 atom. The 6 atoms occupying the center of the faces are shared between 2 cells, so each cell contains 3 atoms. The 4 atoms located completely within the cell, occupying the tetrahedral spaces, are not shared with any other cells. The total is 8 atoms per cell. **3 pt**

4.5. Calculate the density of diamond in g/cm^3 . For reference, consider graphite as having an average density of 2,09–2,23 g/cm^3 .

$$\rho = \frac{m}{V} = \frac{8 \cdot \frac{A_r(C)}{N_A}}{a^3} = \frac{8 \cdot \frac{12,01}{6,022 \cdot 10^{23}}}{(357 \cdot 10^{-12} \cdot 10^2)^3} = 3,51 \text{ g} \cdot \text{cm}^{-3} \quad \mathbf{5 \text{ pt}}$$

In 1985, the family of carbon allotropes called fullerenes was discovered. Their three-dimensional structure is highly variable (spheres, ellipsoids, cylinders, and rings). The IUPAC defines fullerenes as “closed polyhedral cages formed entirely by n trivalent carbon atoms and having 12 pentagonal faces and $(n/2-10)$ hexagonal faces, where $n \geq 20$ ”. This discovery by Kroto, Curl, and Smalley earned them the Nobel Prize a few years later. The first member discovered was C_{60} , which is formed by 12 pentagons and 20 hexagons of sp^2 hybridized carbon atoms. It has a structure identical to a soccer ball and to the geodesic dome designed by the famous Brazilian architect, and for this reason, it is called buckminsterfullerene.

4.6. When Osmani asked Renny how many single and double bonds there are in C_{60} , it was easy for Renny to answer the question, since he was holding the soccer ball they were playing with; he just had to imagine the double bonds. Do you know the answer?

Each carbon atom is sp^2 hybridized, so it forms three single bonds and one double bond with other atoms. But each bond involves two atoms. Therefore, there will be 90 single bonds and 30 double bonds in a C_{60} fullerene molecule. **4 pt** (2 pt each)

4.7. Noticing that Renny had cheated, Osmani got angry and decided to ask a more difficult question: When buckminsterfullerene reacts with oxygen, a small amount of fullerene oxide is produced. This oxide contains 97,83% carbon by mass and is very similar in structure to the starting material. What is the formula of this compound? How many double bonds and how many single bonds does it have?

$$\frac{n(C)}{n(O)} = \frac{\frac{m(C)}{A_r(C)}}{\frac{m(O)}{A_r(O)}} = \frac{\frac{97,83}{12,01}}{\frac{100 - 97,83}{16,00}} \approx \frac{60}{1}$$

Oxide formula: $C_{60}O$

6 pt

(4 pt for calculations and 2 pt for formula)

The formation of the oxide involves the breaking of the weakest bond, which is the π bond of a double bond. Oxygen is divalent and can form two new single bonds with the carbon atoms where the break occurred, making the epoxide group. Therefore, the oxide will have

Simple bonds: 60 (C-C initial) + 1 (C=C broken) + 2 (C-O formed) = 63 **2 pt**

Double bonds: 30 (C=C initial) – 1 (C=C broken) = 29 **2 pt**

Sometime later, the coordinator of the parents' council, Naylen, took the two students to a high-tech center at the Scientific Pole and, with the help of specialists, vaporized 10,0 g of graphite with a laser in an inert atmosphere. Analysis by HPLC chromatography and UV-Vis spectroscopy revealed the presence of the fullerenes C_{60} and C_{70} in a 3:1 molar ratio and a 20% total process yield.

4.8. Propose an expression to estimate the value of the combustion heat ΔH_{comb} of a fullerene C_n as a function of the enthalpy of combustion of carbon and other thermodynamic quantities that you consider appropriate. Why can $\Delta H_{\text{comb}}(C_n)$ be considered directly proportional to n ?

Considering the results in sections 4.6 and 4.7, it is possible to infer the following expression

$$\Delta H_{\text{comb}}^{\text{teo}}(C_n) = n \cdot E(C-C) + 0,5 \cdot n \cdot E(C=C) + n \cdot \Delta H_{\text{comb}}(C) + \Delta H_{\text{sub}}(C_n) \quad \mathbf{8 \text{ pt}}$$

In turn, $\Delta H_{\text{sub}}(C_n)$ must be a function of molecular size, since the sublimation process of a molecular solid involves overcoming Van der Waals forces. Therefore, the final expression must be:

$$\Delta H_{\text{comb}}^{\text{teo}}(C_n) = n \cdot K \quad \mathbf{2 \text{ pt}}$$

4.9. When the entire final mixture resulting from the previous process was combusted in an isothermal calorimeter, 334 kJ were released. Use this data and find the combustion enthalpies of fullerenes C_{60} and C_{70} .

There is excess carbon (graphite) in the mixture, in addition to the fullerenes.

$$n_{\text{total}}(C) = \frac{m_{\text{total}}(C)}{A_r(C)} = \frac{10,0}{12,01} = 0,8326 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

But only 20% of the initial carbon was transformed into fullerenes, therefore

$$n_{\text{fulle}}(C) = 0,20 \cdot 0,8326 = 0,1665 \text{ mol}$$

$$n_{\text{exc}}(C) = 0,80 \cdot 0,8326 = 0,6661 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

$$n_{\text{fulle}}(C) = 60 \cdot n(C_{60}) + 70 \cdot n(C_{70}) = 0,1665 \quad \mathbf{2 \text{ pt}}$$

But as $n(\text{C}_{60}):n(\text{C}_{70}) = 3:1$, then

$$60 \cdot 3 \cdot n(\text{C}_{70}) + 70 \cdot n(\text{C}_{70}) = 0,1665$$

$$n(\text{C}_{70}) = 0,6660 \text{ mmol} \quad \underline{2 \text{ pt}}$$

$$n(\text{C}_{60}) = 3 \cdot n(\text{C}_{70}) = 1,998 \text{ mmol} \quad \underline{2 \text{ pt}}$$

$$Q_{\text{total}} = n(\text{C}_{60}) \cdot \Delta H_{\text{comb}}(\text{C}_{60}) + n(\text{C}_{70}) \cdot \Delta H_{\text{comb}}(\text{C}_{70}) + n_{\text{exc}}(\text{C}) \cdot \Delta H_{\text{comb}}(\text{C}) \quad \underline{2 \text{ pt}}$$

$$-334 = \frac{1,998}{1000} \cdot \Delta H_{\text{comb}}(\text{C}_{60}) + \frac{0,666}{1000} \cdot \Delta H_{\text{comb}}(\text{C}_{70}) + 0,6661 \cdot (-393,5)$$

$$-72 = \frac{1,998}{1000} \cdot \Delta H_{\text{comb}}(\text{C}_{60}) + \frac{0,666}{1000} \cdot \Delta H_{\text{comb}}(\text{C}_{70}) \quad \underline{2 \text{ pt}}$$

According to the data provided on the problem

$$\frac{\Delta H_{\text{comb}}(\text{C}_{60})}{\Delta H_{\text{comb}}(\text{C}_{70})} = \frac{60}{70} \quad \underline{2 \text{ pt}}$$

$$-72 = \frac{1,998}{1000} \cdot \Delta H_{\text{comb}}(\text{C}_{60}) + \frac{0,666}{1000} \cdot \frac{70}{60} \cdot \Delta H_{\text{comb}}(\text{C}_{60})$$

$$\Delta H_{\text{comb}}(\text{C}_{60}) = -25,9 \cdot 10^3 \approx -26 \text{ MJ/mol} \quad \underline{2 \text{ pt}}$$

$$\Delta H_{\text{comb}}(\text{C}_{70}) = -30,3 \cdot 10^3 \approx -30 \text{ MJ/mol} \quad \underline{2 \text{ pt}}$$

The experimental values of $\Delta H_{\text{comb}}^\circ$ for C_{60} and C_{70} are -25930 and -29985 kJ/mol, respectively.

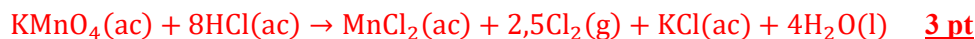
Data: $1 \text{ \AA} = 100 \text{ pm} = 10^{-10} \text{ m}$ $d(\text{pm})$: C-C = 154, C=C = 133, C≡C = 120 $\Delta H_{\text{comb}}^\circ(\text{C}) = -393,5 \text{ kJ/mol}$

Problem 5. Iodine monochloride (15% of total; only 11th and 12th grades)

5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	Total	Total
3	2	4	6	8	22	4	2	7	2	60	15%
3	2	4	6	8	22	4	2	7	2	60	15%

Iodine monochloride (ICl) was discovered by the French chemist and physicist Gay-Lussac in 1814. At room temperature, it is a reddish-brown solid that melts and boils at relatively low temperatures. It can be obtained by passing chlorine gas through solid crystals of diiodine.

5.1. In the Faculty of Chemistry laboratory, Juan Carlos (*Yepe*) obtained dichloromethane *in situ* by oxidizing hydrochloric acid with potassium permanganate. What is the equation for the reaction that occurred?

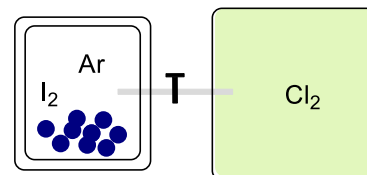


2 pt by the correctly balanced ionic equation

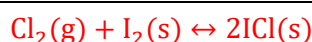
5.2. What essential safety measures should be taken when handling chlorine?

Chlorine is a very toxic gas, and to avoid inhaling it, you should always work under an extraction hood and wear a mask. 2 pt

Juan Carlos likes engineering, so he decided to design an industrial process that could produce ICl in large quantities. To do so, he designed a reactor A (50 L capacity) connected to a reactor B (100 L capacity), both equipped with a device to control temperature and pressure. A private company decided to implement this design in one of its chemical reagent factories. One day, 50,0 mol of solid diiodine were introduced into reactor A under an argon atmosphere, at atmospheric pressure and room temperature. The stoichiometric amount of chlorine required for the reaction was introduced into reactor B, also at room temperature. After opening the stopcock connecting the two reactors, the reaction was allowed to continue for a short time. Finally, the product was recovered from the mixture by distillation, and a yield of 86% was calculated.



5.3. Show that the above reaction occurs to its full extent at room temperature by calculating the equilibrium thermodynamic constant.



$$\Delta G_r^\circ = 2 \cdot \Delta G_f^\circ(\text{ICl}, \text{s}) = 2 \cdot (-14,05) = -28,1 \text{ kJ/mol} \quad \underline{\text{2 pt}}$$

$$K^\circ = \exp\left(-\frac{\Delta G_r^\circ}{R \cdot T}\right) = \exp\left(-\frac{-28,1 \cdot 1000}{8,314 \cdot 298,15}\right) = 8,37 \cdot 10^4 \quad \underline{\text{2 pt}}$$

5.4. Calculate the chlorine pressure inside reactor B before the reaction occurred and the mass of ICl recovered at the end of the process. Assume ideal gas behavior.

$$n(\text{Cl}_2) = n(\text{I}_2) = 50,0 \text{ mol} \quad \underline{\text{1 pt}}$$

$$P(\text{Cl}_2) = \frac{n(\text{Cl}_2) \cdot R \cdot T}{V(\text{B})} = \frac{50 \cdot 8,324 \cdot 298,15}{100} = 1239 \text{ kPa} \quad \underline{\text{2 pt}}$$

$$n_{\text{theo}}(\text{ICl}) = 2 \cdot n(\text{Cl}_2) = 2 \cdot 50 = 100 \text{ mol} \quad \underline{\text{1 pt}}$$

$$M(\text{ICl}) = 126,9 + 35,5 = 162,4 \text{ g/mol}$$

$$n_{\text{exp}}(\text{ICl}) = 0,86 \cdot 100 = 86,0 \text{ mol} = 13966 \text{ g} = 14,0 \text{ kg} \quad \underline{\text{2 pt}}$$

No points are awarded for calculating molar mass.

5.5. Estimate the temperature at which the purification process by distillation is carried out at atmospheric pressure, that is, the standard boiling point of iodine monochloride.



$$\Delta H_{\text{eb}}^{\circ} = \Delta H_{\text{f}}^{\circ}(\text{ICl, g}) - \Delta H_{\text{f}}^{\circ}(\text{ICl, l}) = 17,8 - (-23,93) = 41,7 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

$$\Delta S_{\text{eb}}^{\circ} = S^{\circ}(\text{ICl, g}) - S^{\circ}(\text{ICl, l}) = 247,6 - 135,1 = 112,5 \text{ J/(mol} \cdot \text{K)} \quad \underline{2 \text{ pt}}$$

At the boiling point it is true that $p(\text{ICl}) = 1 \text{ atm}$, $K_{\text{p}}^{\circ} \sim 1$ y $\Delta G_{\text{eb}}^{\circ} = 0$. $\underline{2 \text{ pt}}$

$$T_{\text{eb}} \approx \frac{\Delta H_{\text{eb}}^{\circ}}{\Delta S_{\text{eb}}^{\circ}} = \frac{41,7 \cdot 1000}{112,5} = 371 \text{ K} = 98^{\circ}\text{C} \quad \underline{2 \text{ pt}}$$

The boiling point of ICl reported in the literature is 97°C .

In an attempt to optimize the process, in another experiment, 1.5 times the amount of chlorine required for the reaction was introduced into reactor B, and the reaction was allowed to continue for longer, until equilibrium was established. The operators were surprised when they detected a lower-than-expected final pressure and bright yellow portions in the solid. Furthermore, the final process yield turned out to be lower. They then contacted Juan Carlos for an explanation. After a literature review, the student told them that iodine trichloride had formed: $\text{Cl}_2(\text{g}) + \text{ICl}(\text{s}) \rightarrow \text{ICl}_3(\text{s})$.

5.6. If all reactions proceeded without any kinetic hindrance, calculate in detail the final pressure within the system (reactors A and B connected) and the amount of each product formed.

It is necessary to calculate the equilibrium constant for the new process.



$$\Delta G_{\text{r}}^{\circ} = \Delta G_{\text{f}}^{\circ}(\text{ICl}_3) - \Delta G_{\text{f}}^{\circ}(\text{Cl}_2) - \Delta G_{\text{f}}^{\circ}(\text{ICl, s}) = -22,34 - 0 + 14,05 = -8,29 \text{ kJ/mol} \quad \underline{2 \text{ pt}}$$

$$K^{\circ} = \exp\left(-\frac{\Delta G_{\text{r}}^{\circ}}{R \cdot T}\right) = \exp\left(-\frac{-8,29 \cdot 1000}{8,314 \cdot 298,15}\right) = 28,2 \quad \underline{2 \text{ pt}}$$

Chlorine is in excess, so it will react with some of the ICl formed until equilibrium is reached. The amount left unreacted will be equal to

$$p_{\text{final}}(\text{Cl}_2) = \frac{1}{K_{\text{p}}} = \frac{1}{28,2} = 0,0354 = 3,54 \text{ kPa} \quad \underline{2 \text{ pt}}$$

The argon inside reactor A diffused isothermally towards reactor B, thus reducing the pressure it exerts on the whole system.

$$p_{\text{final}}(\text{Ar}) = \frac{p_{\text{inicial}}(\text{Ar}) \cdot V(\text{B})}{[V(\text{A}) + V(\text{B})]} = \frac{101,3 \cdot 50}{[50 + 100]} = 33,8 \text{ kPa} \quad \underline{2 \text{ pt}}$$

$$\text{Luego, } p_{\text{total}} = p_{\text{final}}(\text{Cl}_2) + p_{\text{final}}(\text{Ar}) = 3,54 + 33,8 = 37,3 \text{ kPa} \quad \underline{2 \text{ pt}}$$

It is necessary to know the amount of chlorine that reacted in the two chemical processes.

$$n_{\text{final}}(\text{Cl}_2) = \frac{p(\text{Cl}_2) \cdot [V(\text{A}) + V(\text{B})]}{R \cdot T} = \frac{3,54 \cdot [50 + 100]}{8,314 \cdot 298,15} = 0,214 \text{ mol} \quad \underline{2 \text{ pt}}$$

The law of mass conservation imposes that

$$n_{\text{inicial}}(\text{I}_2) = 50 = 0,5 \cdot n(\text{ICl}) + 0,5 \cdot n(\text{ICl}_3) \quad \underline{3 \text{ pt}}$$

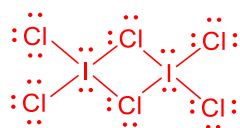
$$n_{\text{inicial}}(\text{Cl}_2) = n_{\text{final}}(\text{Cl}_2) + 0,5 \cdot n(\text{ICl}) + 3/2 \cdot n(\text{ICl}_3)$$

$$1,5 \cdot 50 = 75 = 0,214 + 0,5 \cdot n(\text{ICl}) + 3/2 \cdot n(\text{ICl}_3) \quad \underline{3 \text{ pt}}$$

Solving the equations system we have that

$$n(\text{ICl}) = 75,2 \text{ mol} \quad \underline{2 \text{ pt}} \quad n(\text{ICl}_3) = 24,8 \text{ mol} \quad \underline{2 \text{ pt}}$$

5.7. In the solid state, iodine trichloride exists as a dimer. What is the hybridization and coordination number (number of nearest neighbor atoms) of iodine?



hybridization sp^3d^2 **2 pt** coordination number 4 **2 pt**

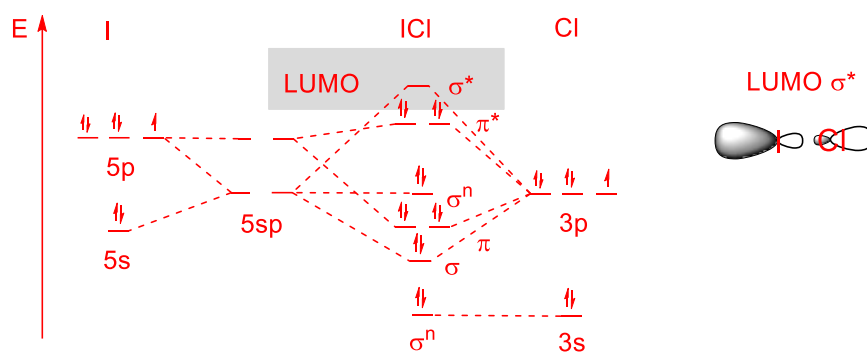
It is not required to draw the molecular structure.

5.8. Juan Carlos knows that ICl is used as an iodinating agent in the synthesis of organic compounds because it has a very electrophilic iodine atom with a high partial positive charge. Why is ICl more reactive than diiodine?

In ICl, the difference in electronegativity between the atoms causes permanent partial charges to exist. In diiodine, the atoms are equal, and only fluctuating charges can exist. **2 pt**

Alternatively, ICl has a weaker bond due to poor overlap between the different-sized orbitals $3p_{Cl}$ and $5p_{I}$, so it reacts more readily.

5.9. The TOM offers an alternative explanation to the previous answer. Construct the molecular orbital energy diagram for ICl, assuming that iodine is sp hybridized. Indicate the LUMO frontier orbital and draw its approximate shape, highlighting which atom it is most developed on.



4 pt for diagram, **1 pt** for identifying LUMO, **2 pt** for drawing orbital

5.10. Why does iodine monochloride, a polar molecule, have a lower boiling point than iodine, a nonpolar molecule?

The most important intermolecular interactions in relatively large molecules are dispersive or London forces (fluctuating dipole-induced dipole). These forces are stronger in molecules composed of large, highly polarizable atoms, such as iodine, than in small, weakly polarizable atoms, such as chlorine. **2 pt**

Thermodynamic data at 25 °C and 100 kPa:

Substance	I ₂ (s)	ICl (s)	ICl (l)	ICl (g)	ICl ₃ (s)	Cl ₂ (g)
ΔH_f°	0	-35,4	-23,93	17,8	-89,5	0
ΔG_f°	0	-14,05	-13,6	-5,5	-22,34	0
S°	116,14	97,93	135,1	247,6	167,4	233,08

Orbital energies (eV):

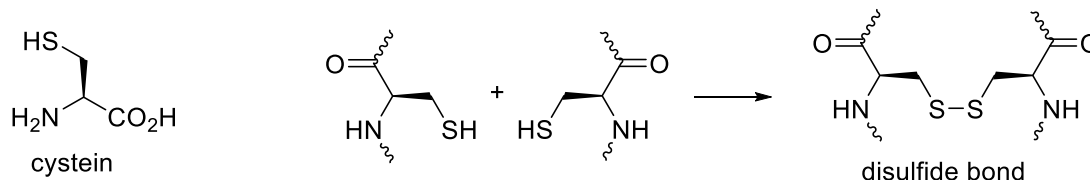
Cl: $3s = -25,23$; $3p = -13,67$ I: $5s = -20,89$; $3p = -11,18$

Problem 6. Hair and chemicals (15% of total; only 12th grade)

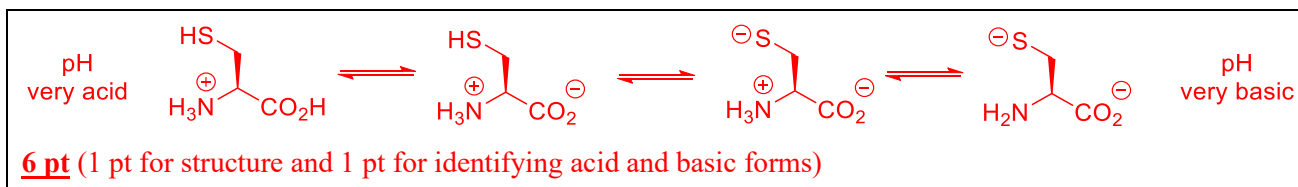
6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	6.10	6.11	6.12	Total	Total
6	4	2	4	10	6	6	2	4	4	8	4	60	15%
6	4	2	4	10	6	6	2	4	4	8	4	60	15%

A very striking example of the effect of chemicals on physical characteristics occurs in hair. Consider Camagüey students Roy and Marco. They both have straight hair, but they differ in color: the former is blond, while the latter is black. Marco's brother, who was already the subject of another question (Carlos Daniel), has curly brown hair. Roy is very interested in organic chemistry and has begun researching the composition of hair proteins. Marco, on the other hand, is more interested in inorganic chemistry and is interested in the reagents that allow hair modification.

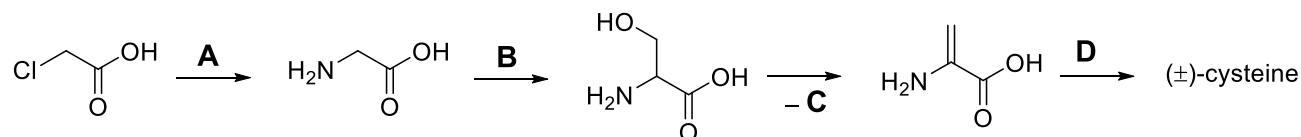
The consistency of hair is determined by the number of disulfide bonds, which give it shape and strength. These bonds are formed between the thiol groups of the amino acid cysteine.



6.1. Amino acids exist in a neutral ionic form called zwitterion. Given the acidity values for cysteine ($pK_{a1} = 1,71$; $pK_{a2} = 8,33$; $pK_{a3} = 10,78$), help Roy to represent all the possible ionic forms it can occur in. Which forms predominate at highly acidic and highly basic pH?



6.2. Roy proposed the following route for obtaining cysteine, from chloroacetic acid. Give the formulas of the compounds A-D.

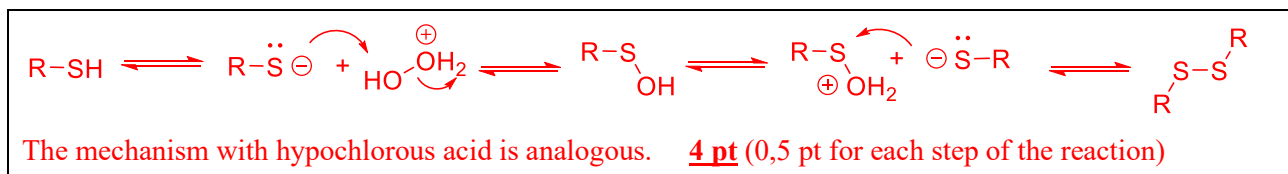


A = NH₃, B = CH₂O, C = H₂O, D = H₂S/NaHS **4 pt** (1 pt each)

6.3. What is the absolute configuration of natural cysteine at the C2 stereogenic center according to the Cahn-Ingold-Prelog notation? Why is cysteine obtained in racemic form in the proposed synthesis?

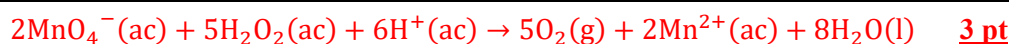
It has R configuration. 1 pt The starting material is an achiral compound. **1 pt**

6.4. Marcos, for his part, discovered that the oxidation of thiol groups to disulfides can be achieved with hydrogen peroxide or hypochlorous acid. Propose the mechanism for one of these reactions.



One day, Marcos decided to get highlights in his hair, and to do so, he asked his mentor for a bottle of diluted aqueous hydrogen peroxide that was in the lab. But Julito reminded him that he had to determine the concentration before using the solution and suggested using a redox titration. Marcos, being very disciplined, designed the following procedure. First, he transferred a 5,00 mL aliquot of the original peroxide into a 100 mL volumetric flask. After adding distilled water and topping it up to the mark, he took a 20,0 mL aliquot of the new solution and titrated it with a 0,0200 mol/L KMnO_4 standard in the presence of sulfuric acid. The values of consumed titrant were 20,0, 19,8, 19,0 and 19,7 mL.

6.5. What is the concentration of the peroxide that Marco found in the laboratory?



The average volume consumed in the titration was 19,8 mL. The value of 19,0 mL should be considered as a gross error and rejected conformingly. $\mathbf{1 \text{ pt}}$

$$c_{\text{diluted}}(\text{H}_2\text{O}_2) = \frac{5}{2} \cdot \frac{c(\text{MnO}_4^-) \cdot V(\text{MnO}_4^-)}{V(\text{H}_2\text{O}_2)} = \frac{5}{2} \cdot \frac{0,0200 \cdot 19,8}{20,00} = 0,0495 \text{ M} \quad \mathbf{4 \text{ pt}}$$

$$c_{\text{laboratory}}(\text{H}_2\text{O}_2) = \frac{100}{5} \cdot 0,0495 = 0,990 \text{ M} \quad \mathbf{2 \text{ pt}}$$

6.6. The optimal peroxide content for treating hair is 5,0% (by mass), with a density of 1,02 g/mL. In contrast, the commercial product has a content of 30% and a density of 1,11 g/mL. What volume of commercial peroxide is needed to prepare one liter of the diluted solution?

$$M(\text{H}_2\text{O}_2) = 2 \cdot 1 + 2 \cdot 16 = 34,0 \text{ g/mol}$$

$$c = \frac{\omega\% \cdot d \cdot 10}{M} \rightarrow c_{\text{comercial}}(\text{H}_2\text{O}_2) = 9,79 \text{ M} \quad c_{\text{diluted}}(\text{H}_2\text{O}_2) = 1,5 \text{ M}$$

$\mathbf{4 \text{ pt}}$ (2 pt for each concentration, no points are given for the molar mass)

$$V_{\text{comercial}}(\text{H}_2\text{O}_2) = \frac{1000 \cdot 1,5}{9,79} \approx 150 \text{ mL} \quad \mathbf{2 \text{ pt}}$$

6.7. After applying the peroxide, Marcos was not satisfied with the result and mistakenly decided to use a more powerful oxidizing agent, hypochlorite, to bleach his hair. Under these conditions, thiols can be oxidized to sulfenic (RSOH), sulfinic (RSO_2H), and sulfonic (RSO_3H) acids. What is the oxidation number of sulfur in these substances? Arrange them in order of increasing acidity and explain your answer.

Oxidation numbers: $\text{RSOH} = 0$, $\text{RSO}_2\text{H} = 2$, $\text{RSO}_3\text{H} = 4$ $\mathbf{3 \text{ pt}}$

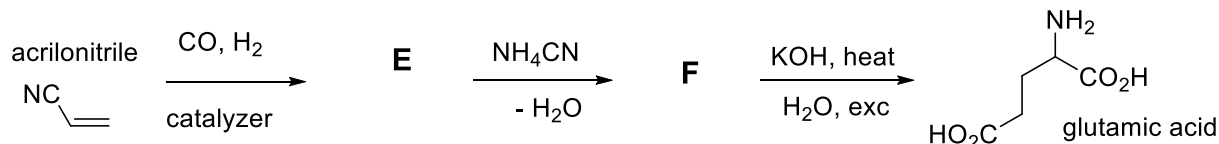
$\text{RSOH} < \text{RSO}_2\text{H} < \text{RSO}_3\text{H}$ $\mathbf{2 \text{ pt}}$

The acidity increases because the sulfur is surrounded by a greater number of oxygen atoms that increase δ^+ on the H/ the conjugated anion has a greater number of resonant structures and is more stabilized. $\mathbf{1 \text{ pt}}$

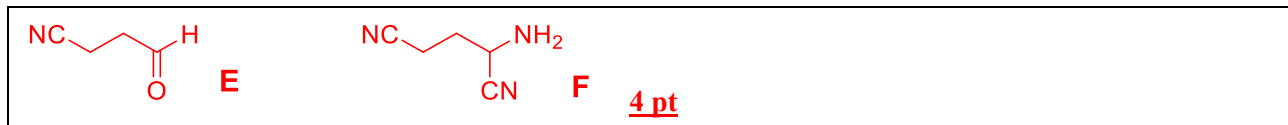
6.8. Hydrogen peroxide is unstable and should be stored at low temperatures and away from light. Formulate the equation for the decomposition reaction of this substance.



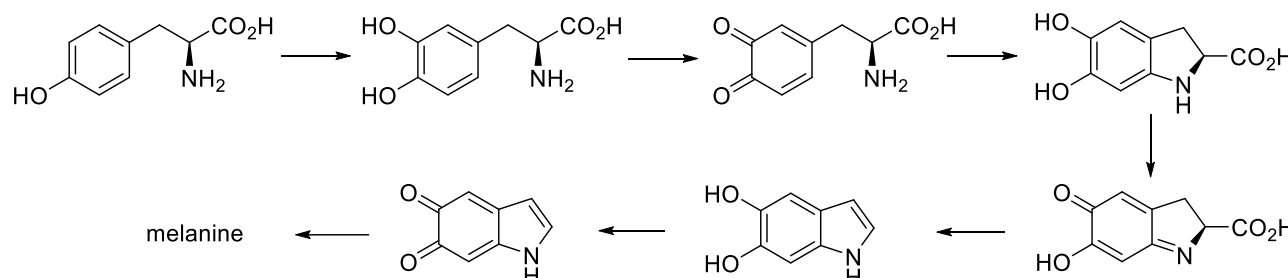
Marcos is very boastful and likes to wash his hair with *shampoo* containing keratin. This protein is rich in sulfur and also in the amino acid glutamic acid, which can be obtained from acrylonitrile.



6.9. Represent the structures of synthetic intermediates E and F.



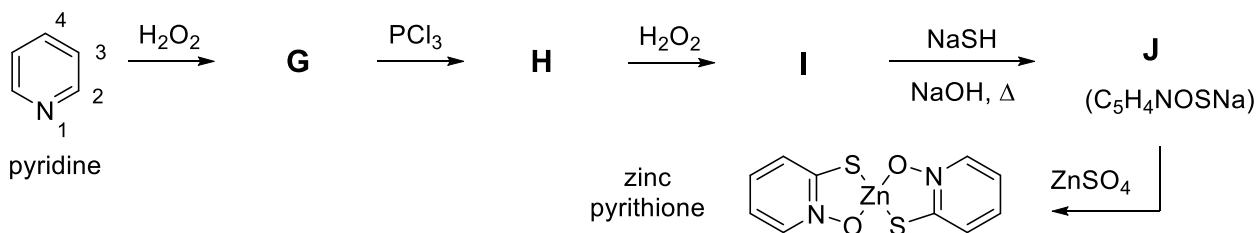
Hair color is determined by the amount of the pigment melanin. Roy knows that melanins are formed by enzyme-catalyzed reactions from the amino acid tyrosine, according to



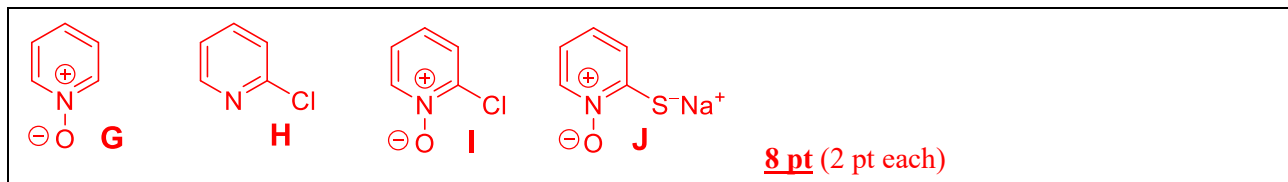
6.10. Answer questions about the above synthetic sequence: i) How many oxidation steps occur? ii) What is the name of the ring-closing reaction? iii) Why is the indole ring formed in the process aromatic?

- i) Four oxidation steps occur. **1 pt**
- ii) Conjugate Michael addition /1,4/ followed by a tautomerization process. **1 pt**
- iii) It is flat, the atoms that form it have sp^2 hybridization, it has 10 π electrons. **2 pt**

To eliminate dandruff, a very annoying fungus that affects the scalp, Roy uses a shampoo whose active ingredient is zinc pyrithione, a synthetic derivative of pyridine.

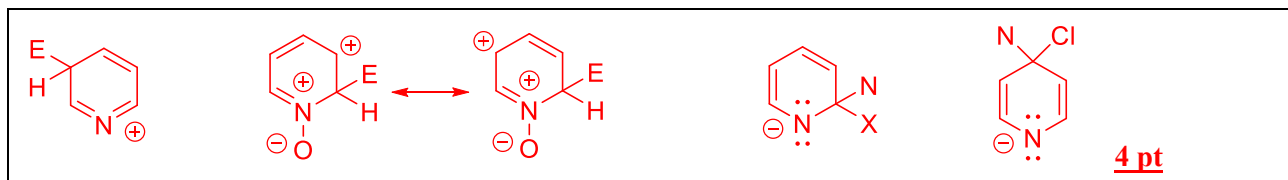


6.11. Represent the structures of the compounds G-J.



Pyridine is an aromatic heterocycle that undergoes S_EAr with strong electrophiles at the 3-position of the ring. In contrast, the corresponding N-oxide **G** favors S_EAr at the 2-position. Halopyridines participate in S_NAr -type reactions more readily when the halogen atoms are located at the 2- and 4-positions (e.g., chloropyridine **H**).

6.12. Represent a relevant (especially stable) structure of the sigma complexes that explain the different regioselectivity in each of the four reactions of S_EAr and S_NAr described above.



Problem 7. Carbon and sulfur (15% of total; 10th, 11th and 12th grades)

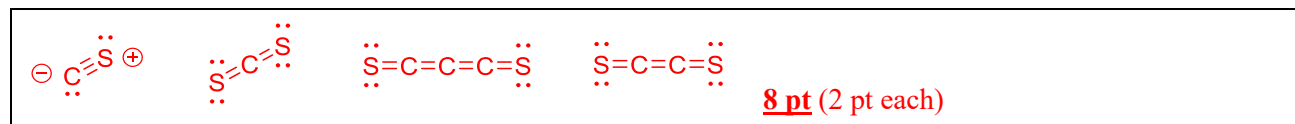
7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
10	8	4	8	2	28	60	15
10	8	4	8	2	28	60	15

Ernesto is a very curious student, and whenever he has the opportunity, he conducts experiments in the school laboratory with technician Ivis and mentor Agustín. After conducting a bibliographic search, the mentor discovered that when carbon and sulfur combine, different binary substances are obtained, and he asked the student to answer the following question.

7.1. What is the formula for the four sulfur compounds of carbon containing approximately 36, 27, and 16% carbon by mass? It is known that in one case, both the monomer and the dimer can exist, and none of the molecules is made up of more than five atoms.

$\frac{n(C)}{n(S)} = \frac{m(C)}{A_r(C)} \div \frac{m(S)}{A_r(S)}$	For $\omega_C = 36\%$	For $\omega_C = 27\%$	For $\omega_C = 16\%$
	$\frac{36}{\frac{12,01}{100 - 36}} \approx 1,5 = \frac{3}{2}$	$\frac{27}{\frac{12,01}{100 - 27}} \approx 1,0 = \frac{1}{1}$	$\frac{16}{\frac{12,01}{100 - 16}} \approx 0,50 = \frac{1}{2}$
2 pt	C_3S_2 2 pt	CS 2 pt	CS_2 2 pt
Since no molecule has more than five atoms, the dimer can only be C_2S_2 . 2 pt			

7.2. Help Ernesto to represent the structures of the four molecules described above.



Ernesto liked the challenge and asked permission to try to obtain the substances at the school. But Ivis denied him, arguing that sulfur compounds are toxic and that the reactions involved were highly exothermic. Fortunately, Agustín has excellent relations with the University of Sancti Spiritus, and they planned to perform the reactions under more controlled conditions the following day.

Upon arriving at the university, Ivis found a **10,0 L** rigid reactor equipped with a pressure gauge and placed it under a fume hood. Ernesto weighed 60,0 g of a carbon-sulfur mixture on a technical balance and placed it inside the container. At that moment, Agustín sent them out for a snack, and when he was left alone in the laboratory, he secretly modified the experimental conditions. Upon returning, the student pressed a button, producing a spark and a small explosion inside the reactor.

After cooling to room temperature, Ernesto was shocked and disappointed because the pressure inside the container was back to where it had been. Upon careful opening, he noticed a very unpleasant odor and the absence of solid residue. Then the mentor confessed his misdeeds: he had removed most of the solid mixture and filled the container with oxygen.

7.3. Can you help Ernesto to formulate the chemical reactions that happened?



“Agustín, you’re always so reckless!” Ivis said angrily and expelled him from the room. Ernesto weighed the portion of the carbon-sulfur mixture his mentor had hidden. He found a mass of only **57,3 g** and poured it completely out of the reactor. The technician made sure to evacuate all the air and replace it this time with the inert gas argon, until a pressure of 1,00 atm was reached. A spark was then ignited, and a strongly exothermic

reaction occurred inside the vessel, resulting in the formation of substances **A** and **B** in a gaseous state. No solid residue was observed through the glass window of the apparatus. The system was then placed in an ice bath. After cooling to 100 °C, the pressure inside the vessel was 3,81 atm. Cooling continued, and upon reaching the temperature of 25 °C, a few drops appeared on the window due to the condensation of some of product **B**, and the pressure dropped to 1,96 atm. Upon reaching 0 °C and after carefully evacuating the gases (only Ar and product **A**), between 37 and 38 mL of a clear, volatile liquid with a very unpleasant odor were recovered (all of product **B**), which, when measured with a hydrometer, had a density of 1,29 g/mL.

7.4. Show that the amount of oxygen initially introduced into the reactor by the mentor was sufficient to react with the small portion of the mixture that was combusted.

Since in both cases the oxidation proceeds to the 4+ state, the maximum amount of oxygen will be required assuming that all the mass that reacted in the first experiment was carbon, which is the element with the lowest molar mass. **2 pt**

$$n_{\max}^{\text{reacc}}(\text{O}_2) = n_{\max}(\text{C}) = \frac{m(\text{oxidized})}{A_r(\text{C})} = \frac{60,0 - 57,3}{12,01} = \frac{2,7}{12,01} = 0,225 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

But the amount of dioxygen introduced into the reactor is greater than the previous mass. **2 pt**

$$n_{\text{reactor}}(\text{O}_2) = \frac{P(\text{O}_2) \cdot V(\text{reactor})}{R \cdot T} = \frac{1,00 \cdot 10,0}{0,08205 \cdot 298,15} = 0,409 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

Vapor pressure is the pressure exerted by the portion of a volatile substance in the gaseous phase, in direct contact—that is, in equilibrium—with the corresponding volatile liquid or solid. Vapor is said to be “saturated” when the gaseous phase contains the largest amount of the volatile substance thermodynamically possible.

7.5. The saturated vapor pressure of **B** at 25 °C is 48,1 kPa. Calculate the maximum amount of **B** in the gas phase that the reactor can hold at that temperature.

$$n_{\text{vap}}^{\max}(\text{B}) = \frac{P_{\text{vap}}^{\circ}(\text{B}) \cdot V(\text{reactor})}{R \cdot T} = \frac{48,1 \cdot 10}{8,3145 \cdot 298,15} = 0,194 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

7.6. Help the student to calculate the identity of **A** and **B**.

The amount of argon inside the reactor must be calculated. The volume occupied by the solid mixture is negligible compared to the volume of the vessel.

$$n(\text{Ar}) = \frac{P(\text{Ar}) \cdot V(\text{reactor})}{R \cdot T} = \frac{1,00 \cdot 10,0}{0,08205 \cdot 298,15} = 0,409 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

At 100 °C the gaseous phase is made up of argon, **A** and **B**.

$$n(\text{total}) = \frac{P_{100^{\circ}\text{C}}(\text{total}) \cdot V(\text{reactor})}{R \cdot T} = \frac{3,81 \cdot 10,0}{0,08205 \cdot 373,15} = 1,244 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

$$n(\text{A}) + n(\text{B}) = n(\text{total}) - n(\text{Ar}) = 1,244 - 0,409 = 0,835 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

At 25 °C The gaseous phase is made up of argon, **A** and a portion of **B**, determined by the vapor pressure.

$$n_{25^{\circ}\text{C}}(\text{gas}) = \frac{P_{25^{\circ}\text{C}}(\text{total}) \cdot V(\text{reactor})}{R \cdot T} = \frac{1,96 \cdot 10,0}{0,08205 \cdot 298,15} = 0,801 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

$$n(\text{A}) + n(\text{Ar}) + n_{\text{vap}}^{\max}(\text{B}) = 0,801 \text{ mol}$$

From the previous section 7.5. we know the amount of **B** that saturates the vapor of the liquid $n_{\text{vap}}^{\max}(\text{B})$. Therefore,

$$n(\text{A}) = 0,198 \text{ mol} \quad \mathbf{2 \text{ pt}} \quad n(\text{B}) = 0,637 \text{ mol} \quad \mathbf{2 \text{ pt}}$$

Note: An alternative form to the above calculation can be:

It is known that $P_{\text{vap}}^{\circ}(\text{B}) = 48,1 \text{ kPa} = 0,475 \text{ atm}$.

$$n(\text{A}) + n(\text{Ar}) = \frac{[P_{25^{\circ}\text{C}}(\text{total}) - P_{\text{vap}}^{\circ}(\text{B})] \cdot V(\text{reactor})}{R \cdot T} = \frac{[1,96 - 0,475] \cdot 10,0}{0,08205 \cdot 298,15} = 0,607 \text{ mol}$$

At 0°C the gas phase is only made up of argon and **A**, while all amount of **B** has condensed.

$$m(\text{B}) = \rho(\text{B}) \cdot V(\text{B}) = 1,29 \cdot 37,5 = 48,4 \text{ g} \quad \underline{\text{2 pt}}$$

But the mass introduced into the reactor for the reaction to occur was

$$m(\text{A}) + m(\text{B}) = 57,3 \text{ g} \quad \underline{\text{2 pt}}$$

$$m(\text{A}) = 57,3 - 48,4 = 8,9 \text{ g} \quad \underline{\text{2 pt}}$$

Both **A** and **B** must contain at least one carbon and one sulfur atom. Therefore,

$$M(\text{A}) = \frac{m(\text{A})}{n(\text{A})} = \frac{8,9}{0,198} = 44,9 \text{ g/mol} \quad \underline{\text{2 pt}}$$

$$\begin{aligned} M(\text{A}) - A_{\text{r}}(\text{C}) - A_{\text{r}}(\text{S}) \\ = 44,9 - 12,0 - 32,1 = 0,8 \text{ g} \cdot \text{mol}^{-1} \end{aligned}$$

$$\text{A} = \text{CS} \quad \underline{\text{3 pt}}$$

$$M(\text{B}) = \frac{m(\text{B})}{n(\text{B})} = \frac{48,4}{0,637} = 76,0 \text{ g/mol} \quad \underline{\text{2 pt}}$$

$$\begin{aligned} M(\text{B}) - A_{\text{r}}(\text{C}) - A_{\text{r}}(\text{S}) \\ = 76,0 - 12,0 - 32,1 = 31,9 \text{ g} \cdot \text{mol}^{-1} \end{aligned}$$

$$\text{B} = \text{CS}_2 \quad \underline{\text{3 pt}}$$

Note: The experiments described in this problem are entirely fictitious and extremely dangerous. **Never** attempt to reproduce them.

Problem 8. Benzoyl peroxide (10% of total; only 10th and 11th grades)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	Total	Total
2	10	4	10	2	4	8	40	10
2	10	4	10	2	4	8	40	10

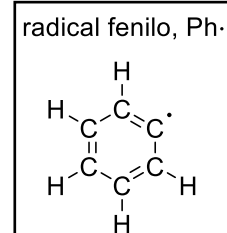
Organic peroxides decompose easily, generating free radicals, which is why they are used as initiators in chain reaction, especially in industrial polymerization processes. One day, Bárbaro and Pedro studied the decomposition of benzoyl peroxide $(\text{PhCOO})_2$ by refluxing in the solvent benzene. These students work well as a team, as their skills complement each other: the former has solid theoretical knowledge, and the latter is very capable in the laboratory.

Pedro's measurements demonstrated that the rate law is first-order with respect to peroxide. However, the values of the reaction rate constants k_{obs} were found to be dependent on the initial concentration of benzoyl peroxide $[\text{PB}]_0$, as shown in the data in the following table.

$[\text{PB}]_0$ (mol/L)	0,0205	0,0830	0,156	0,225	0,283
k_{obs} (10^{-5} s^{-1})	3,85	5,30	6,35	7,10	7,70

These results suggested that, in addition to the unimolecular thermal decomposition of benzoyl peroxide, decomposition processes mediated by some free radical intermediate also occur. Bárbaro found that the following simplified decomposition mechanism is reported in the literature.

1. Initiation: $(\text{PhCOO})_2 \xrightarrow{k_1} 2\text{Ph}\cdot + 2\text{CO}_2$
2. Propagation: $\text{Ph}\cdot + \text{C}_6\text{H}_6 \xrightarrow{k_2} \text{PhC}_6\text{H}_6\cdot$
3. Propagation: $\text{PhC}_6\text{H}_6\cdot + (\text{PhCOO})_2 \xrightarrow{k_3} \text{PhCOOH} + \text{Ph}_2 + \text{Ph}\cdot + \text{CO}_2$
4. Termination: $2\text{PhC}_6\text{H}_6\cdot \xrightarrow{k_4} \text{PhC}_6\text{H}_6\text{C}_6\text{H}_6\text{Ph}$



8.1. Why is the O-O single bond present in peroxides relatively weak?

Oxygen atoms have a high electron density due to their small size, which causes high repulsion and a weak bond. **2 pt**

Bárbaro applied the Steady State Theory to the mechanism and showed that the rates of the initiation step 1 and termination step 4 of the mechanism have to be equal, and that the rate of decomposition of the peroxide obeys a complex expression of the form $v_r = -\frac{d[\text{PB}]}{dt} = (\mathbf{A} + \mathbf{B} \cdot \sqrt{[\text{PB}]}) \cdot [\text{PB}]$, where the constants **a** and **b** can be expressed in terms of the specific rate constants of each step.

8.2. Deduce the complete reasoning that Barbaro followed.

Assuming steady state for the intermediates, their concentrations do not vary over time.

$$d[\text{Ph}\cdot]/dt = 0 = 2 \cdot k_1[(\text{PhCOO})_2] - k_2[\text{Ph}\cdot][\text{C}_6\text{H}_6] + k_3[\text{PhC}_6\text{H}_6\cdot][(\text{PhCOO})_2] \quad \mathbf{2 \text{ pt}}$$

$$d[\text{PhC}_6\text{H}_6\cdot]/dt = 0 = k_2[\text{Ph}\cdot][\text{C}_6\text{H}_6] - k_3[\text{PhC}_6\text{H}_6\cdot][(\text{PhCOO})_2] - 2 \cdot k_4[\text{PhC}_6\text{H}_6\cdot]^2 \quad \mathbf{2 \text{ pt}}$$

Adding both equations and simplifying like terms, we obtain

$$2 \cdot k_1[(\text{PhCOO})_2] = 2 \cdot v_1 = 2 \cdot k_4[\text{PhC}_6\text{H}_6\cdot]^2 = 2 \cdot v_4 \rightarrow v_1 = v_4 \quad \mathbf{2 \text{ pt}}$$

$$[\text{PhC}_6\text{H}_6\cdot] = \sqrt{\frac{k_1}{k_4} \cdot [(\text{PhCOO})_2]} \quad \mathbf{1 \text{ pt}}$$

$$v_r = -d[(\text{PhCOO})_2]/dt = k_1[(\text{PhCOO})_2] + k_3[\text{PhC}_6\text{H}_6\cdot][(\text{PhCOO})_2] \quad \mathbf{1 \text{ pt}}$$

$$v_r = k_1[(\text{PhCOO})_2] + k_3 \sqrt{\frac{k_1}{k_4} \cdot [(\text{PhCOO})_2]} [(\text{PhCOO})_2]$$

$$v_r = \left(k_1 + k_3 \sqrt{\frac{k_1}{k_4}} \cdot [(\text{PhCOO})_2] \right) [(\text{PhCOO})_2] \quad \underline{2 \text{ pt}}$$

The previous expression is of the form $(A + B \cdot \sqrt{[\text{PB}]}) \cdot [\text{PB}]$, as required.

8.3. The values of the specific rate constants are $k_1 = 2,4 \cdot 10^{-5} \text{ s}^{-1}$ and $k_4 = 2,0 \cdot 10^{11} \text{ u}$. State the units of measurement “u” of k_4 and calculate the concentration of the radical $\text{PhC}_6\text{H}_5\cdot$ if we start from an initial peroxide concentration of $0,010 \text{ mol/L}$.

$$\text{XXX} \equiv \text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1} \quad \underline{2 \text{ pt}}$$

$$[\text{PhC}_6\text{H}_5\cdot] = \sqrt{\frac{k_1}{k_4}} \cdot [(\text{PhCOO})_2] = \sqrt{\frac{2,4 \cdot 10^{-5}}{2,0 \cdot 10^{11}}} \cdot 0,010 = 1,1 \cdot 10^{-9} \text{ mol} \cdot \text{L}^{-1} \quad \underline{2 \text{ pt}}$$

8.4. Fit Pedro's results in the table to a straight line and calculate the values of the constants **A** and **B** of Barbaro's rate equation

$[\text{PB}]_0 \text{ (mol/L)}$	0,0205	0,0830	0,156	0,225	0,283	
$k_{\text{obs}} (10^{-5} \text{ s}^{-1})$	3,85	5,30	6,35	7,10	7,70	
$\sqrt{[\text{PB}]_0}$	0,143	0,288	0,395	0,474	0,532	<u>2 pt</u>

Fitting to a straight line k_{obs} versus $\sqrt{[\text{PB}]_0}$ a linearity of $r^2 = 0,999$ is obtained. It is acceptable to draw the graph instead of performing a regression calculation. 4 pt

$$A = \text{intercept} = 2,45 \cdot 10^{-5} \quad \underline{2 \text{ pt}} \quad B = \text{slope} = 9,86 \cdot 10^{-5} \quad \underline{2 \text{ pt}}$$

8.5. When Pedro asked Bárbaro why the fourth reaction step (termination) is the fastest of all, what do you think his answer was?

The termination step is the fastest because it involves the reaction between two radicals, which are very reactive species. 2 pt

8.6. At low concentrations of peroxide $k_{\text{obs}} = k_1$, while at high concentrations of peroxide $k_{\text{obs}} = k_3 \sqrt{\frac{k_1}{k_4}}$. Use the Arrhenius model and express the experimentally observed activation energy for the reaction E_A^{obs} as a function of the activation energies of the individual steps for both cases.

At low concentrations of peroxide, it follows that $E_A^{\text{obs}} = E_{A1}$ 2 pt

At high concentrations of peroxide, it follows that $E_A^{\text{obs}} = \frac{1}{2} E_{A1} + E_{A3} - \frac{1}{2} E_{A4}$ 2 pt

8.7. Estimate the minimum values of the activation energies E_{A1} , E_{A2} , E_{A3} and E_{A4} .

enlace	H-H	C-C	O-O	C-H	C-O	O-H	C=C	C=O
E (kJ/mol)	436	348	146	413	358	463	612	799

$E_{A1} \geq 146 \text{ kJ/mol}$ because it involves breaking the bond O-O. 2 pt

$E_{A2} \geq 612 - 348 = 264 \text{ kJ/mol}$ because it involves breaking the π bond C=C. 2 pt

The third step involves breaking the C-H, C-C, and O-O bonds. The activation energy must be at least sufficient to break the strongest of these bonds, which is C-H. Therefore,

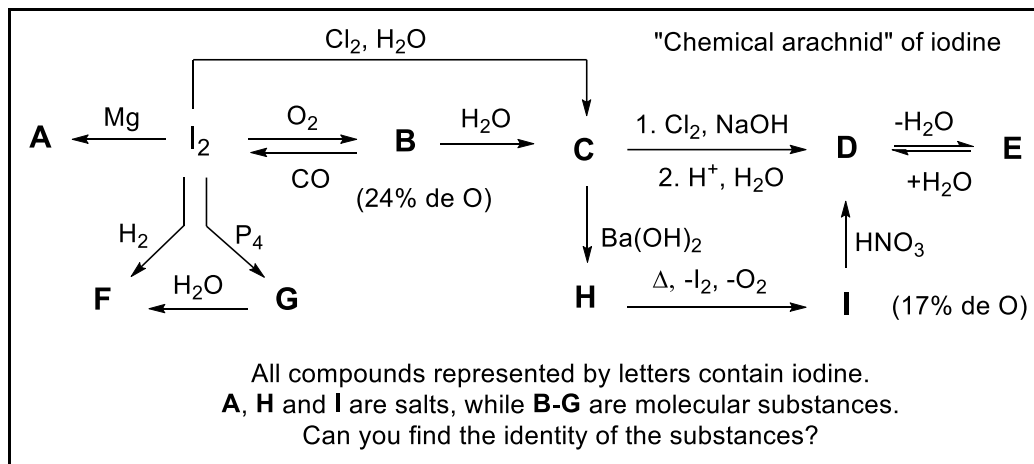
$E_{A3} \geq 413 \text{ kJ/mol}$. 2 pt

$E_{A4} \geq 0 \text{ kJ/mol}$ because no bond is broken in this step. 2 pt

Problem 9. Iodine (15% of total; only 10th and 11th grades)

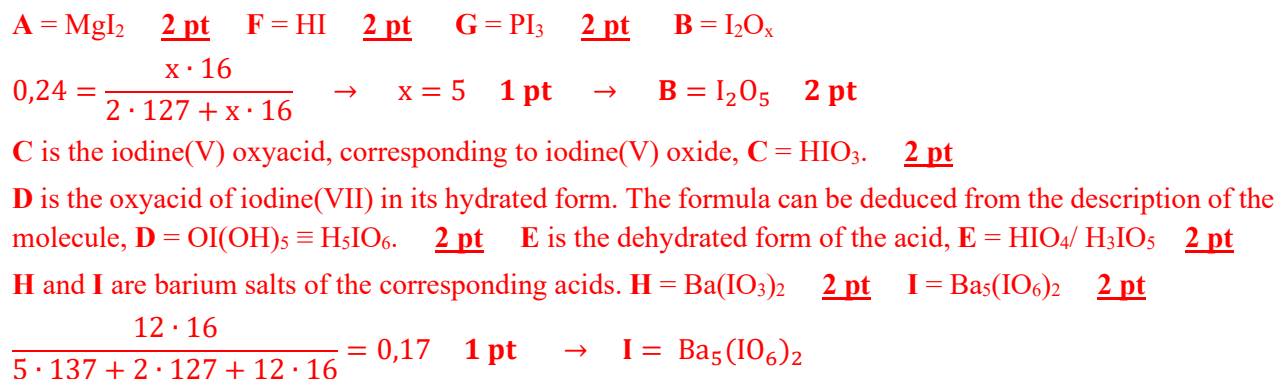
9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	Total	Total
20	11	7	2	2	5	4	4	5	60	15
20	11	7	2	2	5	4	4	5	60	15

When Melissa discovered that Cynthia was afraid of spiders, she came up with a plan to help her overcome her fears. She took a piece of paper that was black on one side and white on the other, wrote a note on the white side, and folded it to create an origami arachnid. Melissa then pretended to be holding a real spider, and Cynthia began to scream in terror –until Melissa confessed–, “Hold it. It won’t bite you; it’s a chemical spider.” After gathering a lot of courage, Cynthia took the origami paper and found the following problem written on the white side of the paper sheet.

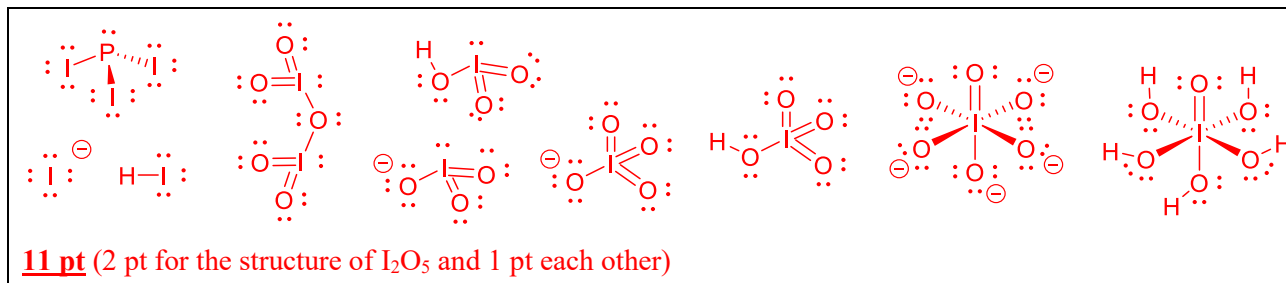


9.1. Help Cynthia to find the formulas for substances A-I. Record your calculations.

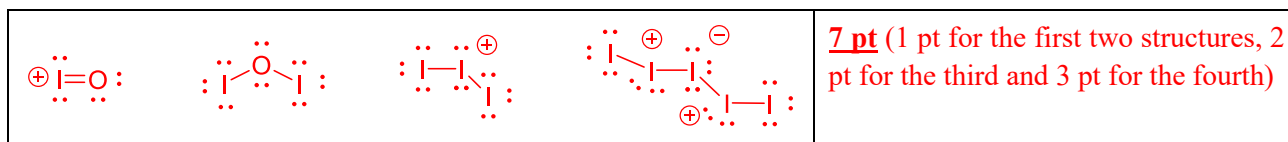
Hint: D and E are iodine oxoacids in the same oxidation state. The molecule of D has a square-base bipyramid geometry, in which five I-O bonds measure 1,87-1,91 Å, and one I-O bond measures 1,78 Å.



9.2. Draw the structures of the iodine species shown in Melissa's note. Note the geometry according to the postulates of Gillespie's theory (TRPEV).



9.3. Melissa discovered in an article that the reaction between diiodine and A in concentrated sulfuric acid is complex and results in the formation of the species IO^+ , I_2O , I_3^+ , I_5^+ . Draw their structures.



9.4. At room temperature, iodine is solid, while bromine is liquid, but chlorine and fluorine are gases. Why is this happening?

Simple halogen substances are molecular. As the atomic number increases, so does the number of electrons and the London dispersion interaction between molecules, so greater thermal energy is required to overcome intermolecular interactions and cause the change of state. **2 pt**

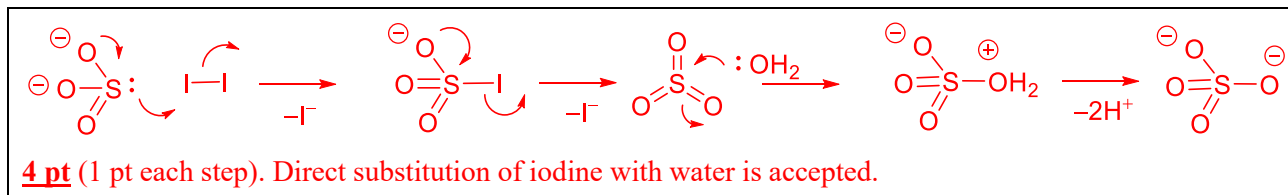
9.5. Hydrogen iodide behaves like a strong acid when dissolved in water, completely dissociating into ions. In contrast, hydrogen fluoride behaves like a weak acid and only partially dissociates. Why is this happening?

The hydrogen atom 1s orbital is very small and can only effectively overlap with the fluorine atom 2p orbitals, so the bond in hydrogen fluoride is stronger and harder to break compared to hydrogen iodide. Alternatively, H^+ is a hard acid and F^- is a hard base; both species form a very strong covalent bond that makes the acid difficult to dissociate. **2 pt**

9.6. Describe the diiodine molecule using Molecular Orbital Theory and state: i) electron distribution, ii) bond order, iii) magnetic properties, and iv) frontier orbitals.

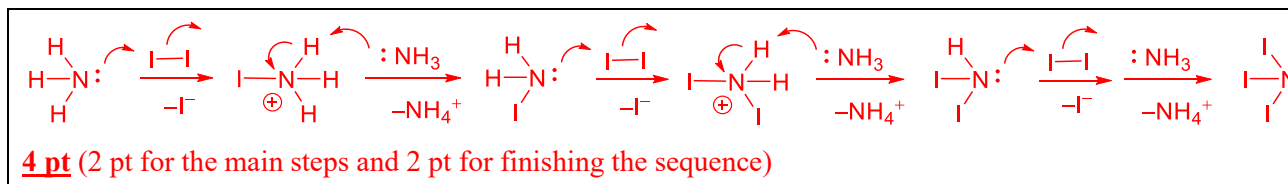
$(\sigma)^2 (\sigma^*)^2 (\sigma)^2 (\pi)^4 (\pi^*)^4 (\sigma^*)^0$ **1 pt** OE = 1 **1 pt** diamagnetic molecule **1 pt**
 HOMO = π^* **1 pt** LUMO = σ^* **1 pt**

9.7. Iodine reacts with the sulfite anion to form sulfate and iodide ions. Propose a probable mechanism for the reaction, indicating the movement of electrons with arrows.

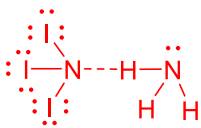


On the day of the science fair, Cynthia decided it was time for revenge. While Melissa was studying unsuspectingly, Cynthia threw some ammonia and iodine in the hallway, right outside the lab entrance, and waited patiently. When the break time bell rang, explosions and screams were heard. It was Melissa, scared, not understanding why the floor was exploding beneath her shoes. And the rest of the chemistry group was laughing out loud, well aware of their mentors' favorite joke. The explosive compound that forms *in situ* during a chemical reaction has the formula $\text{NI}_3 \cdot \text{NH}_3$.

9.8. Propose a probable mechanism for the reaction of nitrogen triiodide formation from ammonia and diiodine, indicating the movement of electrons with arrows.

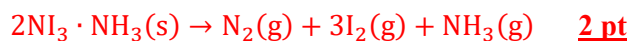


9.9. Draw the structure of nitrogen triiodide ammonate. Explain why this compound is extremely unstable and detonates at the slightest vibration or contact. Write the equation for the decomposition reaction that occurs during the explosion.



2 pt

This compound is unstable because the N-I bonds are weak due to the poor overlap between the $2sp^3$ orbitals of nitrogen and the 5p orbitals of iodine, which have different sizes. 1 pt

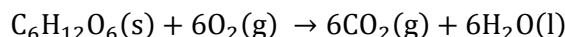


It is accepted to write iodine in solid aggregation state.

Problem 10. Hiking in Gran Piedra (15% of total; only 10th grade)

10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
9	6	4	11	6	4	10	10	60	15
9	6	4	11	6	4	10	10	60	15

Mathematician Ian and physicist Sergio accompanied the chemists who climbed the Gran Piedra (1,225 m above sea level), at the eastern end of the Sierra Maestra mountain range, in Santiago de Cuba, a few weeks ago. The total distance traveled was approximately 30 km, requiring a large expenditure of energy – approximately 2,500 calories. The day before leaving, Ian decided to calculate the amount of food he should take. To do so, he calculated the heat released during the combustion of glucose ($C_6H_{12}O_6$), which occurs at the cellular level and is the main source of energy for the human body:



10.1. How much food (grams of glucose) does a student need to complete the hike?

$$\Delta H_r^\circ = -\Delta H_f^\circ(C_6H_{12}O_6) + 6 \cdot \Delta H_f^\circ(CO_2) + 6 \cdot \Delta H_f^\circ(H_2O) \quad \underline{2 \text{ pt}}$$

$$\Delta H_r^\circ = -(-1273) + 6 \cdot (-394) + 6 \cdot (-286) = -2807 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{2 \text{ pt}}$$

$$M(C_6H_{12}O_6) = 6 \cdot 12,0 + 12 \cdot 1,0 + 6 \cdot 16,0 = 180,0 \text{ g} \cdot \text{mol}^{-1} \quad \underline{1 \text{ pt}}$$

$$n(C_6H_{12}O_6) = \frac{Q}{|\Delta H_r^\circ|} = \frac{2500 \cdot 1000 \cdot 4,184}{2807} = 0,00373 \text{ mol} \quad \underline{2 \text{ pt}}$$

$$m(C_6H_{12}O_6) = 0,00373 \text{ mol} \cdot 180,0 \text{ g} \cdot \text{mol}^{-1} = 0,671 \text{ g} \quad \underline{2 \text{ pt}}$$

10.2. State three reasons why Ian's result is unreliable.

In the body, reactions occur in aqueous solution, so thermodynamic data are required in the aqueous aggregation state.

In addition to glucose and its polymeric forms such as starch, foods contain other carbohydrates and lipids that also constitute a source of energy.

Metabolism is a complex process in which not all the energy released in the oxidation of carbohydrates can be recovered. **6 pt** (1 pt each reason)

On the day of the tour, a meteorological phenomenon known as a trough was over eastern Cuba, causing southwesterly winds loaded with moisture from the Caribbean Sea. As a result, as they approached the summit, just over a kilometer high, the students entered a cloud that remained covering the area all day. Sergio asked the tour guides about the weather and learned that as they left Santiago, on the seashore, the temperature was 1,000 hPa and 25 °C, while at Gran Piedra, both values had dropped to 990 hPa and 15 °C, respectively.

10.3. Consider a 1,00 km³ air mass from the Caribbean Sea moving over Santiago de Cuba toward the mountains. Assuming that there is no diffusion and no mixing with the surrounding air, calculate the volume of the air mass as it ascends to Gran Piedra. Consider ideal behavior.

Considering that the air mass behaves as a closed system, the amount of substance in the gases does not change. **2 pt**

$$V_2 = \frac{V_1 \cdot P_1}{T_1} \cdot \frac{T_2}{P_2} = \frac{1,00 \text{ km}^3 \cdot 1000 \text{ hPa}}{(273 + 25) \text{ K}} \cdot \frac{(273 + 15) \text{ K}}{990 \text{ hPa}} = 0,976 \text{ km}^3 \quad \underline{2 \text{ pt}}$$

Note: The volume occupied by the condensing water vapor is less than 1% and can also be ignored.

If you cannot solve item **10.3**, consider that the volume of the air mass remains 1,00 km³.

Meteorologists use the variable relative humidity (H_r), which is the ratio of the actual pressure exerted by gaseous water (vapor) to the vapor pressure saturating the air at a given temperature. When the air is saturated with water vapor, relative humidity is $H_r = 100\%$. During his time inside the cloud, Sergio wanted to know how much water had condensed in the form of cloud, rain, and dew, as a result of the temperature change. To do this, he used water vapor pressure values (1,70 kPa at 15 °C and 3,17 kPa at 25 °C) and assumed that the air above Gran Piedra was saturated with water vapor, meaning that it contained the maximum possible amount of water in a gaseous state.

10.4. Estimate the maximum amount (L) of precipitation that can be generated from the air mass of 1,00 km³ coming from the Caribbean Sea if it is known that over the city of Santiago de Cuba $H_r = 85\%$.

In Santiago de Cuba, near the sea, $H_r = 85\%$ y $T = 25\text{ °C} = 298\text{ K}$.

$$p(\text{H}_2\text{O})_g = 0,85 \cdot 3,17 = 2,69\text{ kPa} \quad \underline{2\text{ pt}}$$

$$n(\text{H}_2\text{O})_g = \frac{p(\text{H}_2\text{O})_g \cdot V_1}{R \cdot T_1} = \frac{2,69 \cdot 1,00 \cdot 10^{12}}{8,314 \cdot 298} = 1,09 \cdot 10^9\text{ mol} \quad \underline{2\text{ pt}}$$

In Gran Piedra, $H_r = 100\%$ y $T = 15\text{ °C} = 288\text{ K}$.

$$n(\text{H}_2\text{O})_g = \frac{p(\text{H}_2\text{O})_g \cdot V_1}{R \cdot T_1} = \frac{1,70 \cdot 0,976 \cdot 10^{12}}{8,314 \cdot 288} = 6,93 \cdot 10^8\text{ mol} \quad \underline{2\text{ pt}}$$

2 pt for correctly convert from 1 km³ = 10¹² L.

(If the approximate volume of 1,00 km³ is used, the amount of water vapor is 7,10·10⁸ mol.)

The amount of water that precipitates will be the difference from the previous values.

$$n(\text{H}_2\text{O})_{pp} = 1,09 \cdot 10^9 - 6,93 \cdot 10^8 = 4,0 \cdot 10^8\text{ mol} \cdot 18,0\text{ g} \cdot \text{mol}^{-1} = 7,2 \cdot 10^9\text{ g} = 7,2 \cdot 10^6\text{ L} \quad \underline{3\text{ pt}}$$

(If the approximate volume of 1,00 km³ is used, the amount of precipitation is 6,8·10⁶ L.)

If you cannot solve item **10.4**, consider that 5,00 million liters of rainwater fall.

10.5. How much heat (in kJ) was released into the atmosphere as a result of the condensation of water?

$$\Delta H_{\text{vap}}^\circ = -\Delta H_f^\circ(\text{H}_2\text{O}, g) + \Delta H_f^\circ(\text{H}_2\text{O}, l) \quad \underline{2\text{ pt}}$$

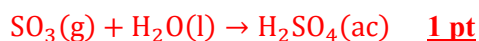
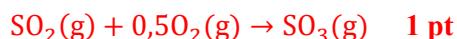
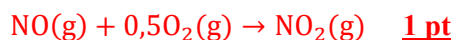
$$\Delta H_{\text{vap}}^\circ = -(-242) + (-286) = -44\text{ kJ} \cdot \text{mol}^{-1} \quad \underline{2\text{ pt}}$$

$$Q = n(\text{H}_2\text{O})_{pp} \cdot \Delta H_{\text{vap}}^\circ = 4,0 \cdot 10^8 \cdot (-44) = -1,8 \cdot 10^{10}\text{ kJ} \quad \underline{2\text{ pt}}$$

(If the approximate volume of 5,0 ML is used, the heat released is 1,2·10¹⁰ kJ.)

During the excursion, Ian learned that acid rain is common on Gran Piedra. This is because the trade winds carry fumes from the Moa industrial zone. This polluted air contains sulfur dioxide and nitric oxide (NO), which, when in contact with oxygen and humidity, generate highly corrosive sulfuric and nitric acids.

10.6. Write the equations corresponding to the formation of acid rain.

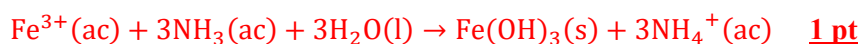
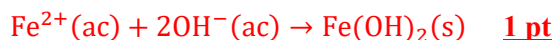
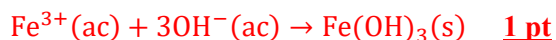
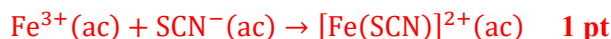


On the way back, Sergio picked up a shiny black stone he found along the way. Upon returning to school and examining the stone, he accidentally discovered that it was attracted to magnets and remained magnetized after the process. He then asked the chemists to analyze the material in the laboratory of the Training Center. The sample dissolved slowly in hot, concentrated hydrochloric acid, leaving no solid residue and creating a yellow-

green solution. A few drops of the solution were then added to test tubes containing 2 cm³ of 0,5 M solutions of potassium rhodanide (sulfocyanide, thiocyanate, KSCN), sodium hydroxide, and ammonia. In the first case, the solution turned deep red, while in the other two, precipitates appeared that changed color until they turned the same brown. The coordinator, Rafael, who has very good connections, requested a quantitative instrumental analysis using ICP and XPS in a specialized laboratory of the Geominera Oriente company. The stone was found to be composed of a single metal oxide with the formula M₃O₄.

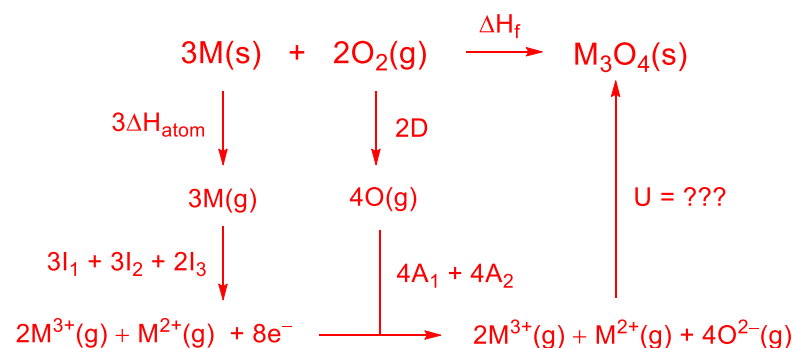
10.7. Identify the metal contained in the stone and write the equations for all the reactions that occurred during the qualitative analysis.

The metal is iron **3 pt**



Other equivalent formulations of the reactions are accepted.

10.8. Calculate the lattice energy of the mixed oxide M₃O₄ from the thermochemical data given. Hint: Note that both M²⁺ and M³⁺ ions are present in the solid.



$$\Delta H_f^\circ(\text{M}_3\text{O}_4) = 3\Delta H_{\text{atom}}^\circ(\text{M}) + 3I_1(\text{M}) + 3I_2(\text{M}) + 2I_3(\text{M}) + 2D(\text{O}_2) + 4A_1(\text{M}) + 4A_2(\text{M}) + U(\text{M}_3\text{O}_4)$$

8 pt (1 pt for each correct energy contribution)

$$-1117 = 3 \cdot 415 + 3 \cdot 759 + 3 \cdot 1561 + 2 \cdot 2957 + 2 \cdot 498 + 4 \cdot (-141) + 4 \cdot 845 + U(\text{M}_3\text{O}_4)$$

$$U(\text{M}_3\text{O}_4) = -19048 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2\ pt}$$

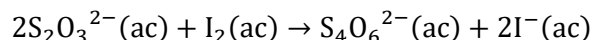
Calculation with Kapustiskii's equation using $\bar{z}(\text{Fe}^{n+}) = 2,67$ and $\bar{r}(\text{Fe}^{n+}) = 72 \text{ pm}$ gives -18300 kJ/mol .

1 cal = 4,184 J $\rho(\text{H}_2\text{O}) = 1,00 \text{ g/mL}$ 1 bar = 100 kPa = 1000 hPa = 0,987 atm 1 L = 1 dm³
 ΔH_f° (kJ/mol, 25°C): C₆H₁₂O₆(s) = -1273, CO₂(g) = -394, H₂O(l) = -286, H₂O(g) = -242, M₃O₄ = -1117
D(O₂) = 498 kJ/mol, A₁(O) = -141 kJ/mol, A₂(O) = 845 kJ/mol, I₁(M) = 759 kJ/mol, I₂(M) = 1561 kJ/mol, I₃(M) = 2957 kJ/mol; $\Delta H_{\text{atomization}}^\circ(\text{M}) = 415 \text{ kJ/mol}$.

Problem 11. Iodometry or iodimetry (20% of total; only 11th and 12th grades)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	11.11	11.12	11.13	11.14	11.15	11.16	Total	Total
2	4	2	2	6	2	7	2	8	4	8	4	9	4	9	7	80	20
2	4	2	2	6	2	7	2	8	4	8	4	9	4	9	7	80	20

Mentor Orestes (better known by his surname Landrove) knows that iodine methods are suitable for performing quantitative determinations of certain analytes. Although it is possible to perform the titration directly using iodine, he always recommends using an alternative titrant, typically sodium thiosulfate, which is a mild reducer capable of reacting with the iodine formed by an analyte. The main reaction that occurs is



11.1. One of the reasons why iodine is not preferred as a titrant is that it is very volatile. To counteract this difficulty, it is common to add excess iodide ions. Why?

Iodine reacts with iodide to form triiodide, which is a stable and soluble complex. **2 pt**

11.2. Calculate the equilibrium constant of the main reaction of the titration at 25 °C.

$$\Delta E_r^\circ = E^\circ(\text{I}_2/\text{I}^-) - E^\circ(\text{S}_4\text{O}_6^{2-}/\text{S}_2\text{O}_3^{2-}) = 0,54 - 0,08 = 0,46 \text{ V} \quad \mathbf{2 \text{ pt}}$$

$$K_e = \exp\left(\frac{z \cdot F \cdot \Delta E_r^\circ}{R \cdot T}\right) = \exp\left(\frac{2 \cdot 96485 \cdot 0,46}{8,314 \cdot 298}\right) = 3,6 \cdot 10^{15} \quad \mathbf{2 \text{ pt}}$$

11.3. The reaction above is based on the high affinity of the sulfur in thiosulfate for iodine. Explain this phenomenon according to the Hard and Soft Acids and Bases Theory.

Iodine and sulfur are large, polarizable atoms with a low charge density, so they are classified as soft, having a high affinity for forming covalent bonds. **2 pt**

11.4. The stability of thiosulfate solutions is very sensitive to the pH of the medium. For example, acidic impurities cause cloudiness and a pungent odor. Write the equation for the reaction that occurs.



During the training prior to the International Chemistry Olympiad, Landrove prepared several experiments based on diiodine titrations for Hector. First, they attempted to determine the vitamin C (ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$, abbreviated as AA hereafter) content in the juice of various fruits. Since ascorbic acid (AA) is not generated through any human metabolic pathway, consuming foods rich in vitamin C, such as citrus fruits, tomatoes, and potatoes, is necessary to prevent diseases like scurvy, which is characterized by collagen deficiency. Hector decided to quantify AA using a redox titration with a standard iodine solution, forming iodide and dehydroascorbic acid ($\text{C}_6\text{H}_6\text{O}_6$). Starch is used as a visual indicator, which forms a deep blue compound in the presence of traces of iodine.

11.5. What masses of iodine and potassium iodide does Hector need to weigh to prepare one liter of aqueous solution with molar concentrations of 0,00100 M and 0,0500 M, respectively?

$$M(\text{I}_2) = 2 \cdot 126,9 = 253,8 \text{ g} \cdot \text{mol}^{-1} \quad \mathbf{1 \text{ pt}} \quad M(\text{KI}) = 39,1 + 126,9 = 166,0 \text{ g} \cdot \text{mol}^{-1} \quad \mathbf{1 \text{ pt}}$$

$$m(\text{I}_2) = 253,8 \text{ g} \cdot \text{mol}^{-1} \cdot 0,00100 \text{ mol} \cdot \text{L}^{-1} \cdot 1,00 \text{ L} = 0,254 \text{ g} \quad \mathbf{2 \text{ pt}}$$

$$m(\text{KI}) = 166,0 \text{ g} \cdot \text{mol}^{-1} \cdot 0,0500 \text{ mol} \cdot \text{L}^{-1} \cdot 1,00 \text{ L} = 8,30 \text{ g} \quad \mathbf{2 \text{ pt}}$$

11.6. Formulate the equation for the reaction between iodine and ascorbic acid.



11.7. In general, the pH of the I_2 solution is adjusted to avoid disproportionation of this species. Which medium is preferred? Justify numerically.

In an acidic medium, iodine can disproportionate to HOI and I^- . **1 pt**

$$\Delta E_r^\circ = E^\circ(I_2/I^-) - E^\circ(HOI/I_2) = 0,54 - 1,45 = -0,91 \text{ V} < 0 \quad \text{1 pt}$$

This reaction is not thermodynamically favored.

In a basic medium, iodine can disproportionate to OI^- and I^- . **1 pt**

$$E^\circ(OI^-/I_2) = [2 \cdot E^\circ(OI^-/I^-) - 1 \cdot E^\circ(I_2/I^-)]/1 = 2 \cdot 0,49 - 0,54 = 0,44 \text{ V} \quad \text{1 pt}$$

$$\Delta E_r^\circ = E^\circ(I_2/I^-) - E^\circ(OI^-/I_2) = 0,54 - 0,44 = 0,10 \text{ V} > 0 \quad \text{1 pt}$$

This reaction is thermodynamically favored.

It is preferable to perform titrations with iodine in an acidic medium so that undesired disproportionation does not occur. **2 pt**

Landrove macerated a variety of fruits and vegetables separately in distilled water using a mortar and pestle. He then filtered the mixture, washed the solid portion, and transferred the collected liquid to a 50-mL flask, topping up with additional filtrated liquid to the required volume. Hector then transferred 10,0 mL of the solution to an Erlenmeyer flask using a volumetric pipette. He added approximately 20 mL of distilled water and 2 mL of 1% starch and titrated with the 0,00100 M iodine standard solution until the first sign of a permanent blue color appeared. The results are shown in the table.

Food	orange	plump	pepper	guava	parsley
$V(I_2)$, mL	12,5	9,8	16,2	20,0	8,6

11.8. What glassware is needed to perform a gravity filtration?

Universal support, smooth funnel, ring with clamp, filter paper, collection flask. **2 pt**

11.9. Typically, AA content ranges from 10-30 mg per 100 mL of orange juice. Do Hector's results correspond to the values reported in the literature?

$$M(AA) = 6 \cdot 12,0 + 8 \cdot 1,0 + 6 \cdot 16,0 = 176,0 \text{ g} \cdot \text{mol}^{-1} \quad \text{1 pt}$$

By stoichiometry, the amounts of iodine and AA that react are equal. **1 pt**

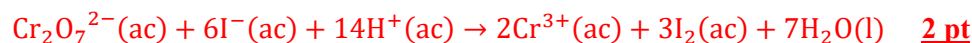
$$n(I_2) = c(I_2) \cdot V(I_2) = 0,00100 \text{ mol} \cdot \text{L}^{-1} \cdot 12,5 \text{ mL} = 0,0125 \text{ mmol} \quad \text{2 pt}$$

$$m(AA) = 0,0125 \text{ mmol} \cdot 176,0 \text{ g} \cdot \text{mol}^{-1} = 2,20 \text{ mg en } 10 \text{ mL} = 22,0 \text{ mg en } 100 \text{ mL} \quad \text{2 pt}$$

Hector's results are within the range of reported values. **2 pt**

Landrove then explained to Hector that the concentration of the standard solution he had prepared was probably incorrect and suggested using back titration to determine the ascorbic acid content. These methods generate an excess of iodine by reacting with an oxidizing primary standard. Some of the iodine formed reacts with the analyte, and the excess is titrated with thiosulfate.

11.10. Formulate the reactions corresponding to the reactions of iodide ions with dichromate and iodate ions in an acidic medium.



11.11. The guava juice that Hector analyzed contained 35,2 mg of AA per 100 mL. Considering that only AA contributes to the acidity of the food, estimate the pH of the solution.

$$c(\text{AA}) = \frac{m(\text{AA})}{M(\text{AA}) \cdot V(\text{D})} = \frac{35,2 \text{ mg}}{176,0 \text{ g} \cdot \text{mol}^{-1} \cdot 100 \text{ mL}} = 2,00 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{2 pt}}$$

Taking into account the degree of dilution of the sample and the values of the constants, it is possible to ignore the second dissociation of the acid in the calculations. 1 pt

$$K_{a1} = \frac{[\text{H}^+][\text{C}_6\text{H}_7\text{O}_6^-]}{[\text{C}_6\text{H}_8\text{O}_6]} \approx \frac{[\text{H}^+]^2}{c_0(\text{C}_6\text{H}_8\text{O}_6) - [\text{H}^+]} \rightarrow 10^{-4,1} = \frac{[\text{H}^+]^2}{0,002 - [\text{H}^+]} \quad \underline{\text{2 pt}}$$

Solving the quadratic equation gives $[\text{H}^+] = 3,61 \cdot 10^{-4} \text{ M}$ and $\text{pH} = 3,4$. 3 pt

A penalty of -2 pt is imposed if acid dissociation is not considered in the equilibrium.

At the end of the session, the mentor tested the student's knowledge of chemical equilibrium. Landrove explained that the copper(II) ion is not capable of oxidizing iodide ions, as evidenced by the standard electrode potential values. However, the mentor dissolved some copper sulfate pentahydrate crystals in pure water in a test tube. Upon adding a portion of potassium iodide and stirring, the solution immediately changed color, first to green and then to dark yellow, and a precipitate appeared. After filtering and washing with water, the precipitate turned white.

11.12. Help Hector explain why the redox reaction of copper(I) iodide precipitate formation occurs, taking into account the polarization of ions and the Hard and Soft Acids and Bases Theory.

The Cu(II) ion has a high charge, a relatively small radius, and an incomplete configuration, making it highly polarizing. The iodide ion is very large, with numerous electrons, and is highly polarizable. The polarization between the two ions results in a net electron transfer. The formation of the CuI precipitate is favored because the Cu(I) and iodide ions are soft acids and bases, respectively, and can form bonds with significant covalent character. 4 pt

11.13. Hector found it easy to demonstrate numerically that in the presence of excess iodide ions, the process described above is enhanced. Can you do it?



$$\Delta G_3^{\circ} = -1 \cdot 96485 \cdot (0,15 - 0,54) - 2,303 \cdot 8,314 \cdot 298 \cdot 11,9 = -3,03 \cdot 10^4 \text{ J} < 0 \quad \underline{\text{2 pt}}$$

$$K_e = \exp\left(\frac{\Delta G_3^{\circ}}{-R \cdot T}\right) = \exp\left(\frac{-3,03 \cdot 10^4}{8,314 \cdot 298}\right) = 2,0 \cdot 10^5 \quad \underline{\text{1 pt}}$$

11.14. Calculate the solubility (in mol/L) of copper(I) iodide in water.

The only relevant equilibrium is the dissociation of the salt. $\text{CuI}(\text{s}) \rightleftharpoons \text{Cu}^{+}(\text{ac}) + \text{I}^{-}(\text{ac})$

Taking into account the charge and mass balance equations, it is concluded that the solubility is

$$c_{\text{total}}(\text{CuI}) = [\text{Cu}^{+}] = [\text{I}^{-}] = \sqrt{K_{\text{ps}}} = \sqrt{10^{-11,9}} = 1,1 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{4 pt}}$$

11.15. Calculate the solubility (mol/L) of copper(I) iodide in a 0,50 mol/L aqueous solution of potassium iodide. Consider the following additional equilibrium that is established



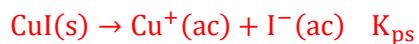
In this case, two reactions must be considered. The equilibrium concentration of iodide ions is assumed to be equal to the initial concentration because the formation of the complex from the precipitate occurs to a small extent. **1 pt**

$$c_{\text{total}}(\text{CuI}) = [\text{Cu}^+] + [\text{CuI}_2^-] = [\text{Cu}^+] + K_f \cdot [\text{Cu}^+] \cdot [\text{I}^-]^2 = \frac{K_{\text{ps}}}{[\text{I}^-]} + K_f \cdot K_{\text{ps}} \cdot [\text{I}^-] \quad \mathbf{6 \text{ pt}}$$

$$c_{\text{total}}(\text{CuI}) = \frac{10^{-11,9}}{0,50} + 7,1 \cdot 10^8 \cdot 10^{-11,9} \cdot 0,50 = 4,5 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1} \quad \mathbf{2 \text{ pt}}$$

Despite the common ion effect, solubility increases due to the formation of the complex ion.

Alternatively, one can directly state that $c_{\text{total}}(\text{CuI}) \approx [\text{CuI}_2^-] = K_f \cdot K_{\text{ps}} \cdot [\text{I}^-]$.



11.16. Explain why copper(I) ions are unstable in aqueous solution using electrode potential values (quantitative) and the Theory of Hard and Soft Acids and Bases (qualitative).

$$1 \cdot E^\circ(\text{Cu}^+/\text{Cu}) = 2 \cdot E^\circ(\text{Cu}^{2+}/\text{Cu}) - 1 \cdot E^\circ(\text{Cu}^+/\text{Cu}) = 2 \cdot 0,34 - 0,15 = 0,53 \text{ V} \quad \mathbf{2 \text{ pt}}$$



$$\Delta E_r^\circ = E^\circ(\text{Cu}^+/\text{Cu}) - E^\circ(\text{Cu}^{2+}/\text{Cu}) = 0,53 - 0,15 = 0,38 \text{ V} > 0 \quad \mathbf{2 \text{ pt}}$$

Since the reaction potential is positive and relatively large, the equilibrium constant will also be large, so the direct disproportionation reaction will be favored. **1 pt**

The Cu(II) ion is an intermediate-hard acid that is more stabilized by electrostatic interaction with water molecules (hard base), compared to the Cu(I) ion, which is a soft acid. **2 pt**

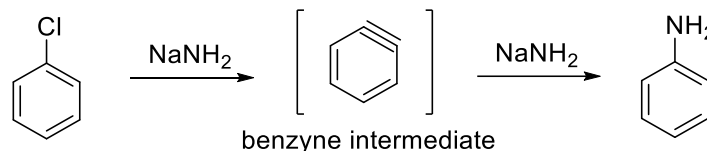
The dissociation constants of ascorbic acid are $\text{pK}_{\text{a1}} = 4,1$ y $\text{pK}_{\text{a2}} = 11,8$ $\text{pK}_{\text{ps}}(\text{CuI}) = 11,9$

$E^\circ(\text{V})$: $\text{S}_4\text{O}_6^{2-}/\text{S}_2\text{O}_3^{2-} = 0,08$; $\text{I}_2/\text{I}^- = 0,54$; $\text{IO}^-/\text{I}^- = 0,49$; $\text{HIO}/\text{I}_2 = 1,45$; $\text{Cu}^{2+}/\text{Cu}^+ = 0,15$; $\text{Cu}^{2+}/\text{Cu} = 0,34$

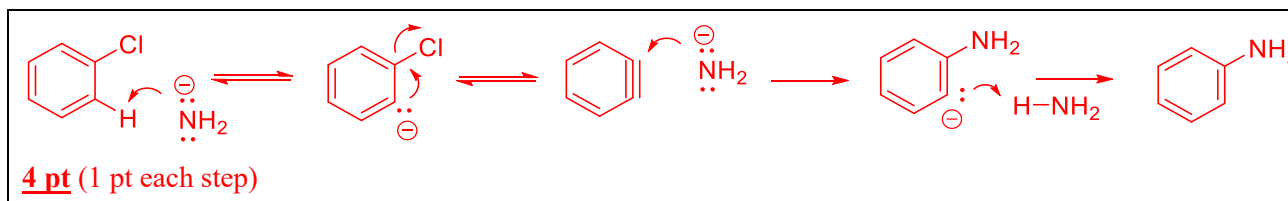
Problem 12. Impossible molecules (20% of total; only 12th grade)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	12.9	12.10	12.11	12.12	12.13
4	4	6	12	2	2	2	6	2	6	2	3	3
4	4	6	12	2	2	2	6	2	6	2	3	3
12.14	12.15	12.16	12.17	12.18	12.19	12.20	12.21	12.22	12.23	12.24	Total	Total
2	2	2	3	2	6	2	2	3	2	---	80	20
2	2	2	3	2	6	2	2	3	2	---	80	20

Part I. Benzene derivatives can be converted into a reactive intermediate called benzyne. One way to form a benzyne is the treatment of chlorobenzene with NaNH_2 . In this case, the benzyne forms aniline under the same conditions.

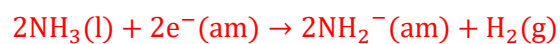
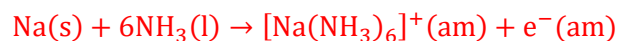


12.1. Propose a mechanism for the formation of aniline from chlorobenzene.



Professor Rolando (also known as *Rolo* to his friends) is the most senior member of the National Pre-Selection and his passion is Organic Chemistry. A student once asked *Rolo* to obtain sodium amide to synthesize aniline. Under strictly anhydrous conditions, in an inert glove compartment, *Rolo* introduced a tiny bit of sodium into liquid nitrogen at -33°C . An intense blue color was immediately obtained, but only after adding a pinch of iron(III) nitrate was intense bubbling observed and the solution began to discolor.

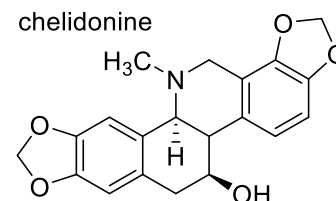
12.2. Formulate chemical equations that explain the above results.



4 pt (2 pt each step). 2 pt if only the total equation is given .

Then, a student asked *Rolo* if benzenes had any other applications besides obtaining aniline. It turns out that benzenes can also be formed into complex molecules, and *Rolo* discovered that a benzyne intermediate is part of the key step for the total synthesis of the natural product chelidonine. The synthetic sequence in question is discussed below.

Starting material **A** is oxidized first with OsO_4 and HIO_4 . This combination results in the same oxidation as neutral ozonolysis, giving **B**. **B** is further oxidized in hydrolyzing conditions to provide **C**. **C** is brominated under acidic conditions to give monobrominated compound **D**. A sequence of silylating reagent HMDS, butyllithium, and triflic anhydride (Trf_2O) result in benzyne precursor **G**. **G** is converted into the reactive benzyne **H** using CsF .

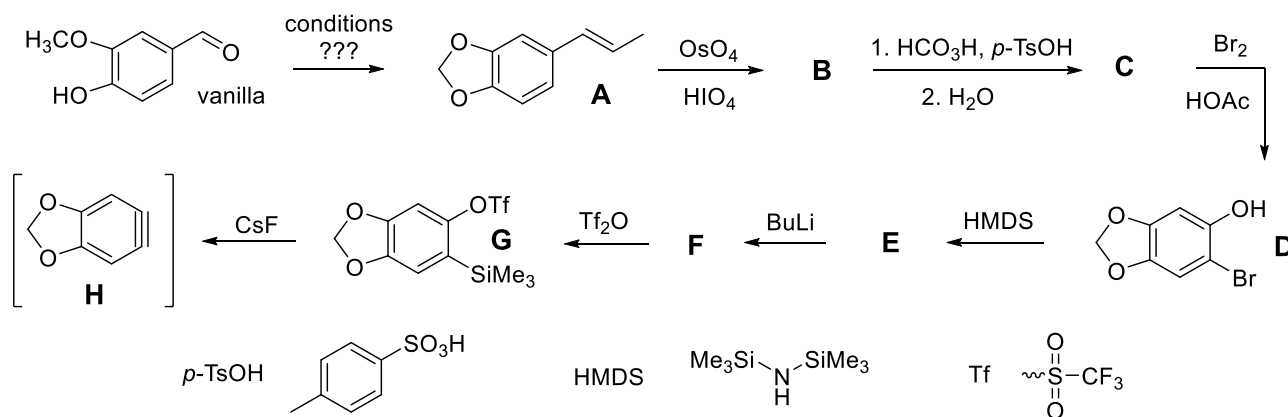


12.3. Propose a sequence that allows obtaining **A** from the natural product vanillin.

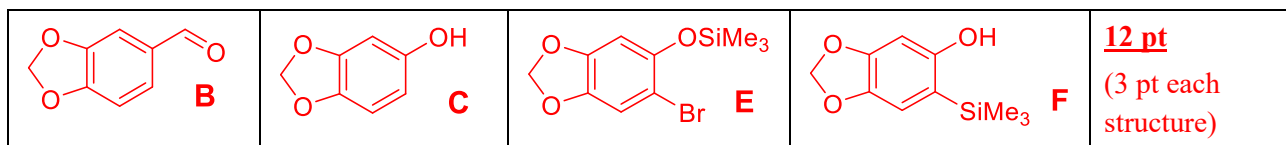
Several answers are possible, for example,

1. HI , H_2O ; 2. $\text{Ph}_3\text{P}=\text{CHCH}_3$ exc; 3. CH_2I_2 , NaH

1. $\text{Ph}_3\text{P}=\text{CHCH}_3$ exc; 2. BBR_3 , H_2O ; 3. CH_2I_2 , K_2CO_3 **6 pt** (2 pt each reagent)



12.4. Draw the structures of B, C, E and F.



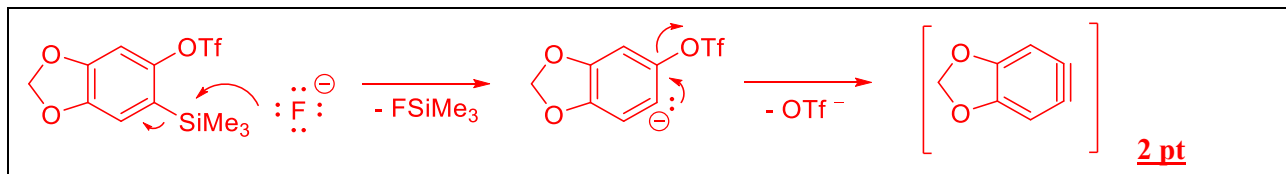
12.5. Give the name of the oxidation from B to C.

Baeyer-Villiger's reaction. **2 pt**

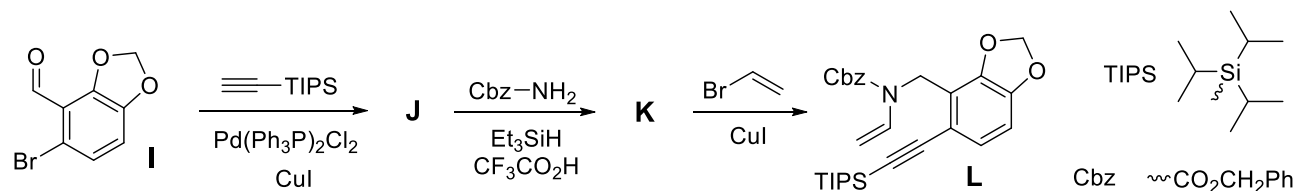
12.6. Explain the regioselectivity of monobromination to give D.

The carbon atom where the reaction occurs is the most activated because it is in the *ortho* position relative to the phenol group, *para* relative to the OR group and it is the position with least steric hindrance. **2 pt**

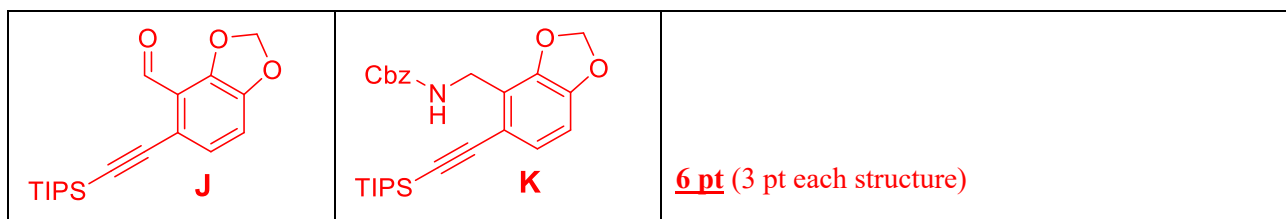
12.7. Propose a mechanism for the formation of benzyne H from G.



The benzyne reacts with a separate compound L. The synthesis of L proceeds as shown below. A Sonogashira coupling with I gives J. Treatment of J with CbzNH₂ in the presence of Et₃SiH and trifluoroacetic acid gives K. Another metal-catalyzed coupling gives the reaction partner L.



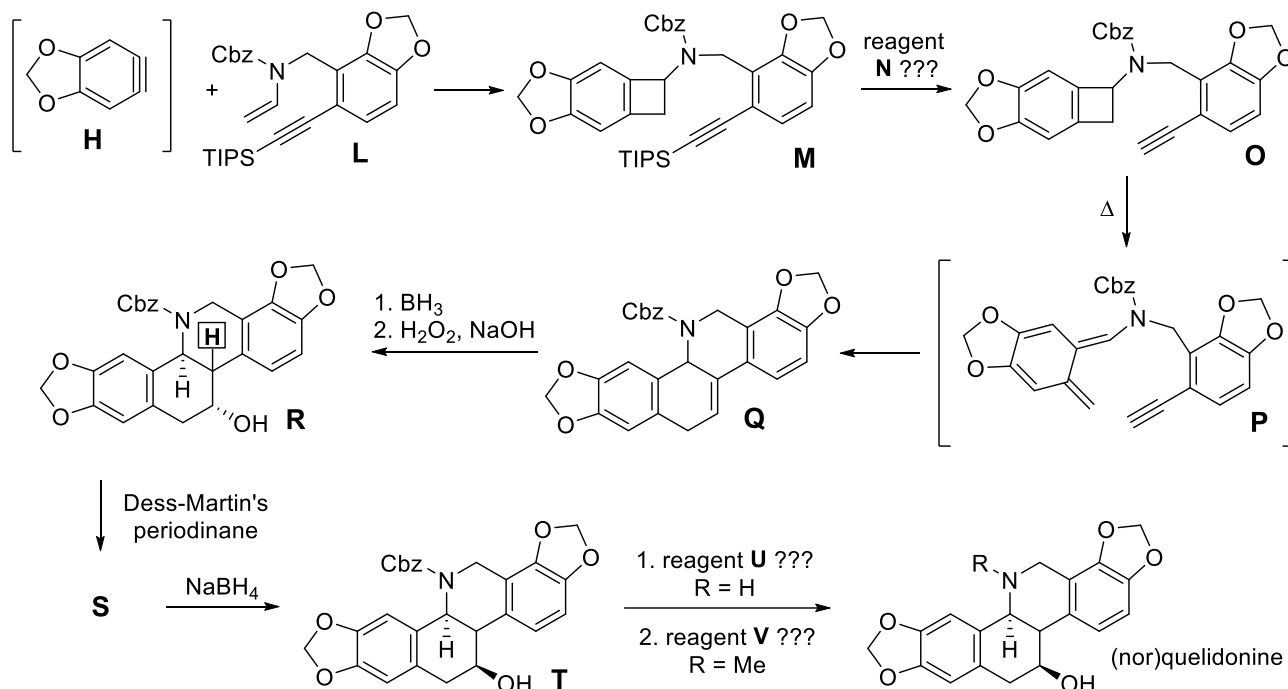
12.8. Draw the structures of J and K.



12.9. Mention the role of Et_3SiH in the conversion of **J** to **K**.

Reducing agent and promoting the release of water. **2 pt**

Next, *in situ* formed benzyne **H** and molecule **L** undergo a cascade to give a polycyclic system **M**. The final steps of the synthesis convert **Q** into norchelidonine. Borane followed by basic H_2O_2 gives alcohol **R**. Treatment with Dess-Martin periodinane converts **R** into **S**, which is followed by reaction with NaBH_4 to provide **T**. A single step converts **T** into the natural product norchelidonine and a second step in chelidonine.



12.10. Propose the reagents **N**, **U** and **V**.

N = CsF / TBAF ; **U** = H_2 , Pd-C / H_2 , Ni-Raney ; **V** = CH_3I , Na_2CO_3 / CH_2O , NaBH_3CN **6 pt** (2 pt each)

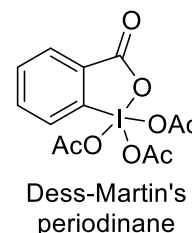
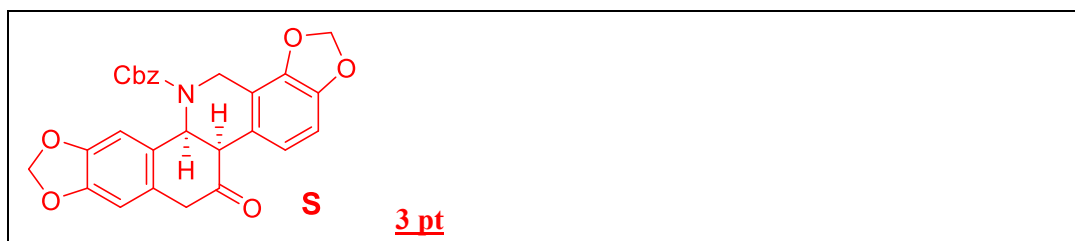
12.11. Give the name of the reaction that converts **P** into **Q**.

Diels-Alder's reaction. **2 pt**

12.12. Draw **R** showing the stereochemistry of the hydrogen atom highlighted inside a square.



12.13. Draw the structure of **S**.



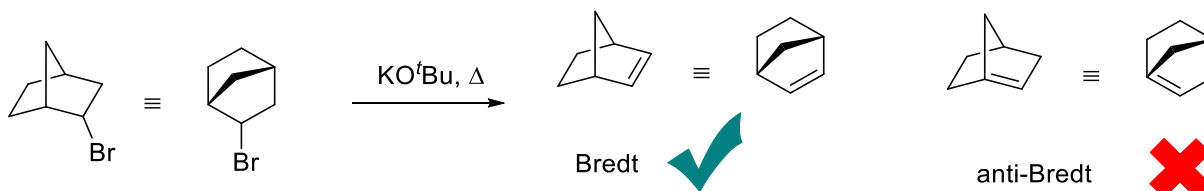
12.14. What is the function of the Dess-Martin periodinane?

Oxidizing agent. **2 pt**

12.15. What isomeric relationship exists between **R** and **T**?

R are diastereoisomers. **2 pt**

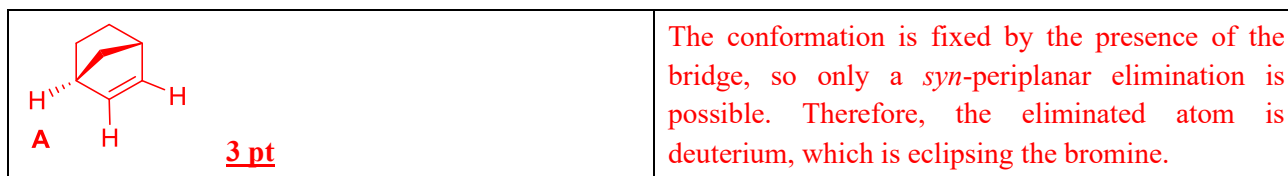
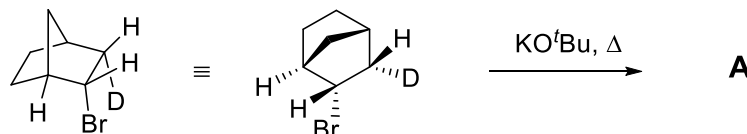
Part II. When Rolo explains bridged bicycles to his students, he always teaches them Bredt's rule: "A double bond cannot be located in the bridgehead position unless the rings are large enough", typically with eight or more members. Elimination reactions in these compounds generally obey Bredt's rule, as illustrated below.



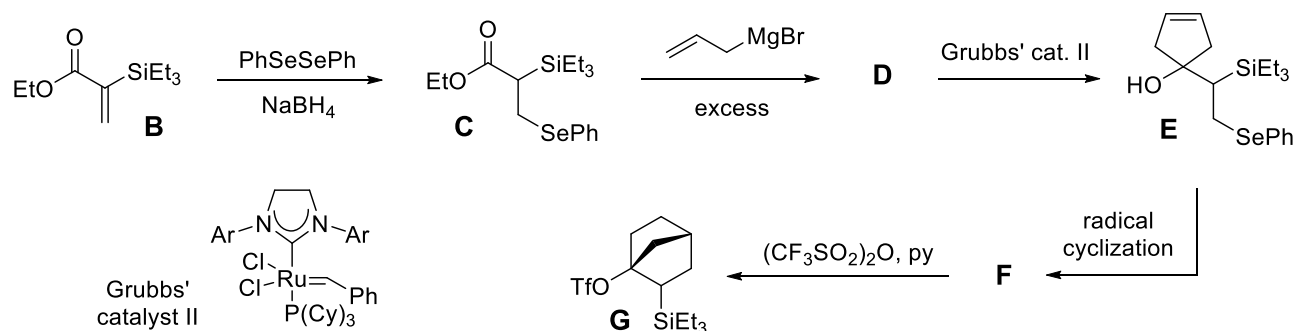
12.16. What is the cause of Bredt's rule?

The high ring tension resulting from the deviation from the ideal 120° angle of the alkene. **2 pt**

12.17. Consider the elimination in the following isotopically labeled compound. Predict the product of the reaction, clearly indicating the isotopes present in the product.



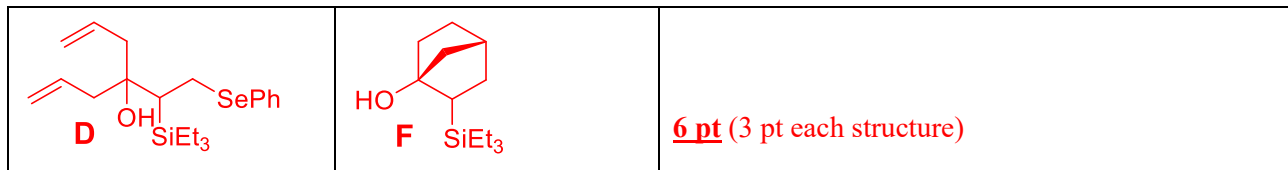
To Rolo's surprise, in 2024, McDermott and coworkers developed a way to prepare the anti-Bredt alkene. Selenium is added to enone **B** to provide **C**. Treatment with excess Grignard reagent gives compound **D**, which can undergo olefin metathesis with Grubbs catalyst to provide cyclic compound **E**. **E** undergoes radical cyclization to give compound **F**, which is converted to **G** using TiF_2O in the presence of pyridine.



12.18. State whether selenides are hard or soft nucleophiles. Explain your answer and explain how it affects the reaction from **B** to **C**.

Selenides are soft nucleophiles that give rise to conjugate addition in enones, due to the softer electrophilic position (C4). **2 pt**

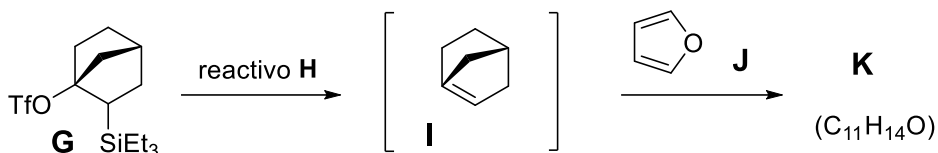
12.19. Draw the structures **D** and **F**.



12.20. What is the oxidation number of ruthenium in Grubbs' catalyst?

2+ **2 pt**

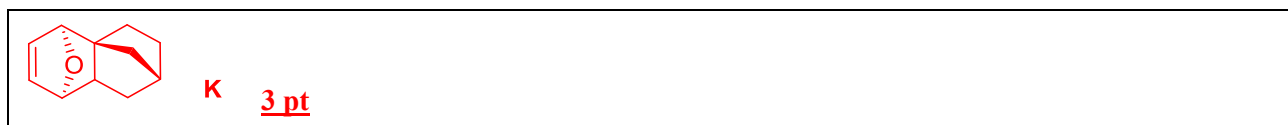
G can be converted into the anti-Bredt alkene **I**; however, **I** is extremely reactive so the researchers provided a reaction partner **J**, so that **I** could react immediately.



12.21. Suggest a reagent **H** that allows obtaining the anti-Bredt alkene.

H = CsF / TBAF / NaOH / LiBu **2 pt** HF is also acceptable.

12.22. Represent the structure of the final product **K**.



12.23. In fact, **G** exists as two diastereomers (**G1** and **G2**), and only one of them can be converted into the anti-Bredt alkene. Based on the structural information provided, predict which isomer is able to form the alkene. Justify your answer.



The **G2** isomer can form the alkene because it has a conformation close to *syn*-coplanar. **2 pt**